

# Triangles: their kinds and their elements

Three sides, three angles, and the two ways of classifying every triangle there is.

LESSON 46

1 × 40 MIN

MATEMATIKA 7, §19

S8 8.1

LESSON CLOCK

10'

12'

12'

6'

|                         |        |
|-------------------------|--------|
| Elements and notation   | 10 min |
| Two classifications     | 12 min |
| The triangle inequality | 12 min |
| Homework                | 6 min  |

LEARNING OBJECTIVES

- Name the elements of a triangle and use the standard notation.
- Classify triangles by their sides and by their angles.
- Use the triangle inequality to decide whether three lengths can form a triangle.
- Identify the median, the altitude and the bisector of a triangle.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 118–122        |
| Cambridge | Stage 8, pp. 78–84 |
| Workbook  | Exercise 8.1       |

01

## Explanation

### Elements and notation

A triangle has three **vertices**, three **sides** and three **angles**. The standard notation names the side by the small letter of the opposite vertex.

| ELEMENT   | WRITTEN                        |
|-----------|--------------------------------|
| vertices  | $A, B, C$                      |
| sides     | $a = BC, b = CA, c = AB$       |
| angles    | $\angle A, \angle B, \angle C$ |
| perimeter | $P = a + b + c$                |

Three special segments from a vertex:

| SEGMENT  | GOES TO                              |
|----------|--------------------------------------|
| median   | the midpoint of the opposite side    |
| altitude | the opposite side, at a right angle  |
| bisector | the opposite side, halving the angle |

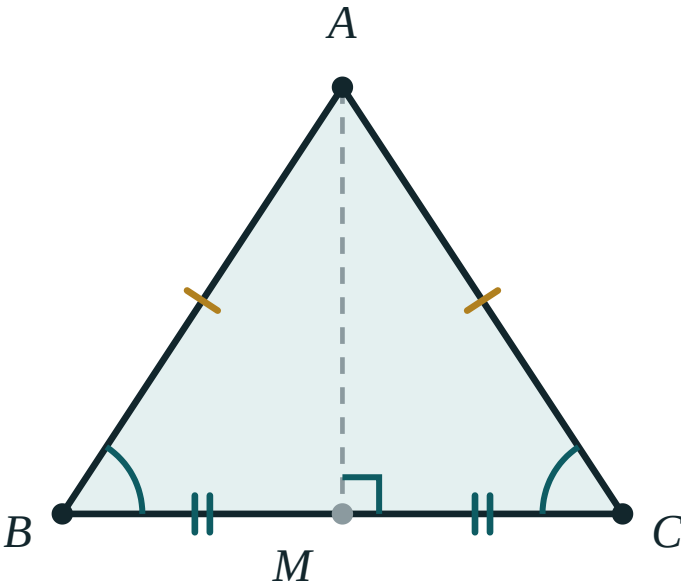
SIDE A FACES ANGLE A

That single convention makes every later formula readable — the sine rule, the cosine rule, the area formula. It is worth fixing now.

Two classifications

| BY SIDES    | MEANS               | BY ANGLES | MEANS            |
|-------------|---------------------|-----------|------------------|
| scalene     | all sides different | acute     | all angles acute |
| isosceles   | two sides equal     | right     | one angle 90°    |
| equilateral | all three equal     | obtuse    | one angle obtuse |

The two classifications are independent: a triangle may be isosceles and right, or scalene and obtuse, and so on. An equilateral triangle is always acute, with all angles 60°.



$AB = AC \implies \angle B = \angle C$

An isosceles triangle — two equal sides and two equal base angles.

EVERY EQUILATERAL TRIANGLE IS ISOSCELES

“Two sides equal” does not mean “exactly two”. The equilateral triangle is the special case in which all three are.

The triangle inequality

Each side is less than the sum of the other two

$$a < b + c, b < c + a, c < a + b$$

It is enough to check the **longest** side: if the longest is less than the sum of the other two, all three inequalities hold.

| LENGTHS   | CHECK         | TRIANGLE? |
|-----------|---------------|-----------|
| 3, 4, 5   | $5 < 3 + 4$   | yes       |
| 2, 3, 6   | $6 > 2 + 3$   | no        |
| 4, 4, 8   | $8 = 4 + 4$   | no — flat |
| 7, 10, 15 | $15 < 7 + 10$ | yes       |

EQUALITY GIVES NO TRIANGLE

4, 4, 8 lays the two short sides flat along the long one. The inequality must be strict.

## 02 Worked examples

**EXAMPLE 1** Can a triangle have sides 2, 3, 6? And 7, 10, 15?

- ① Longest of the first: 6.
- ②  $2 + 3 = 5 < 6$  — no triangle.
- ③ Longest of the second: 15.
- ④  $7 + 10 = 17 > 15$  — yes.

ANSWER

No; yes

**EXAMPLE 2** A triangle has  $a = 5$  and  $b = 9$ . Between which values must  $c$  lie?

- ①  $c < 5 + 9 = 14$
- ②  $9 < 5 + c \Rightarrow c > 4$
- ③ Combine.
- ④  $4 < c < 14$

ANSWER

$4 < c < 14$

**EXAMPLE 3** Classify a triangle with sides 5, 5, 8 and one angle of  $106^\circ$ .

- ① Two sides equal: isosceles.  
By sides.
- ② One angle above  $90^\circ$ : obtuse.  
By angles.
- ③ Both classifications apply at once.
- ④ An obtuse isosceles triangle.

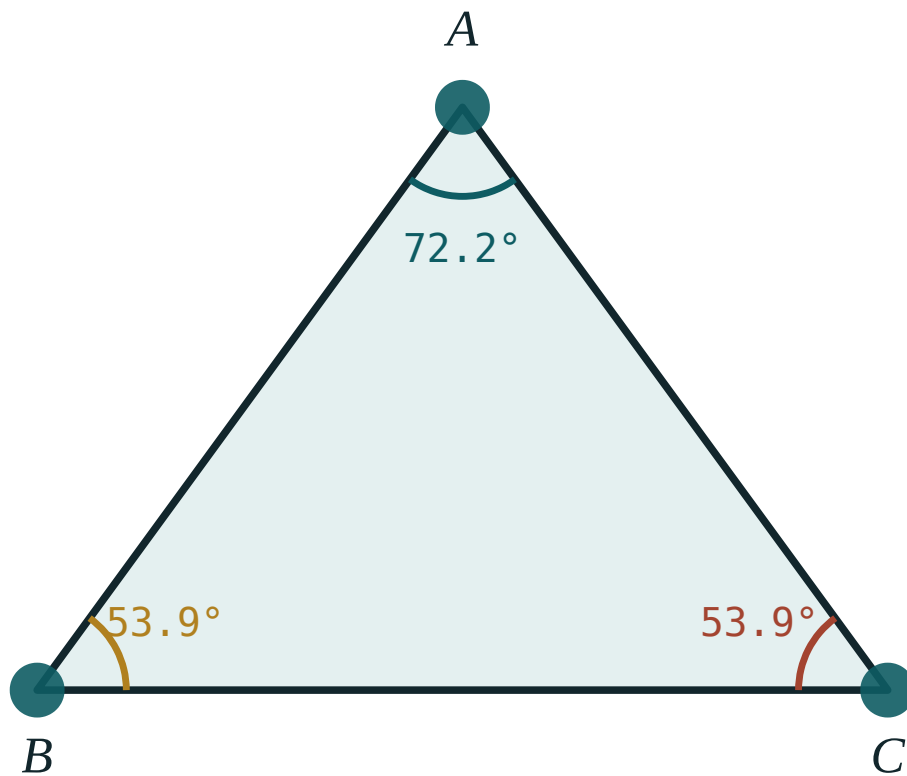
**ANSWER**

Obtuse isosceles

**03** Interactive model

Give out three strips of card of lengths 2, 3 and 6 cm and ask the class to make a triangle; the failure is more convincing than the inequality.

## Kinds of triangle

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side  $BC$ largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH     | O'ZBEKCHA     | РУССКИЙ        |
|-------------|---------------|----------------|
| Triangle    | Uchburchak    | Треугольник    |
| Vertex      | Uchi          | Вершина        |
| Side        | Tomon         | Сторона        |
| Scalene     | Turli tomonli | Разносторонний |
| Isosceles   | Teng yonli    | Равнобедренный |
| Equilateral | Teng tomonli  | Равносторонний |
| Median      | Mediana       | Медиана        |
| Altitude    | Balandlik     | Высота         |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Side  $a$  is opposite:

A triangle with two equal sides is:

An equilateral triangle has angles:



Can 2, 3, 6 form a triangle?

yes

no

sometimes

only if right

A median goes to:

the midpoint

at a right angle

the bisected angle

the vertex

With  $a = 5$ ,  $b = 9$ ,  $c$  satisfies:

$c < 14$

$c > 4$

$4 < c < 14$

$c = 14$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Side opposite  $\angle B$

ANSWER  $b$

2. A triangle with all sides different

ANSWER Scalene

3. A triangle with all sides equal

ANSWER Equilateral

4. Angles of an equilateral triangle

ANSWER  $60^\circ$  each

5. Can 3, 4, 5 be a triangle?

ANSWER Yes

6. Can 2, 3, 6 be a triangle?

ANSWER No

7. Can 4, 4, 8 be a triangle?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Can 7, 10, 15 be a triangle?

ANSWER Yes

2.  $a = 5$ ,  $b = 9$ : range of  $c$

ANSWER  $4 < c < 14$

3.  $a = 6$ ,  $b = 6$ : range of  $c$

ANSWER  $0 < c < 12$

4. Classify 5, 5, 8 with a  $106^\circ$  angle

ANSWER Obtuse isosceles

5. Classify 3, 4, 5

ANSWER Right scalene

6. Perimeter of a triangle 7, 9, 12

ANSWER 28

7. A median goes to

ANSWER The midpoint

**Hard · stretch and reason**

7 PROBLEMS

1. An isosceles triangle with perimeter 20 and base 8

ANSWER Sides 6, 6, 8

2. The possible integer values of  $c$  with  $a = 5$ ,  $b = 9$

ANSWER 5 to 13

3. How many of them?

ANSWER 9

4. An isosceles triangle with sides 3 and 7

ANSWER Only 3, 7, 7

5. Why not 3, 3, 7?

ANSWER  $3 + 3 < 7$

6. A triangle with perimeter 12 and integer sides: how many?

ANSWER 3

7. Name them

ANSWER 2,5,5; 3,4,5; 4,4,4

## 07 Homework

### Homework — 5 tasks

Check the longest side against the sum of the other two, every time.

- Can a triangle have sides 4, 6, 11? And 8, 9, 16?
- A triangle has  $a = 7$  and  $b = 11$ . Give the range of  $c$ .
- Classify a triangle with sides 6, 6, 6.
- Classify a triangle with a  $90^\circ$  angle and two equal sides.
- An isosceles triangle has perimeter 26 and base 10. Find the equal sides.

# The first criterion of congruence of triangles

Two sides and the angle between them fix a triangle completely.

LESSON 47–48

2 × 40 MIN

MATEMATIKA 7, §20

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

|                       |        |
|-----------------------|--------|
| What congruence means | 13 min |
| The criterion         | 23 min |
| Writing a proof       | 22 min |
| Consequences          | 18 min |
| Homework              | 4 min  |

LEARNING OBJECTIVES

- Define congruent triangles and use the symbol  $\cong$ .
- State and apply the SAS criterion.
- Write a proof of congruence in statement-and-reason form.
- Deduce equal sides and angles from a congruence.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 123–129        |
| Cambridge | Stage 8, pp. 48–56 |
| Workbook  | Exercise 5.1       |

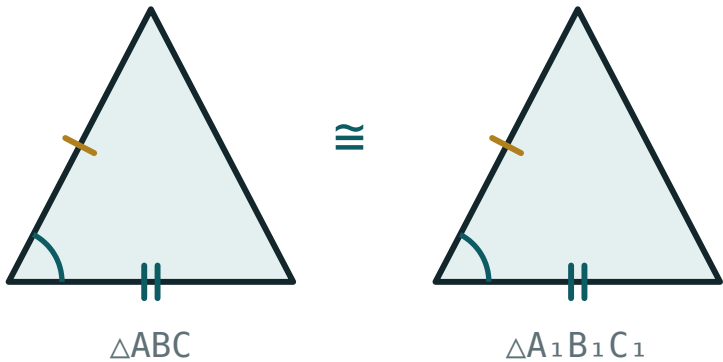
01

## Explanation

### What congruence means

Two triangles are **congruent** if one can be laid exactly on the other. Written  $\triangle ABC \cong \triangle A_1B_1C_1$ , with the vertices in matching order.

Congruent triangles have **six** pairs of equal elements: three sides and three angles.



Congruent triangles — same shape and same size, in any position.

THE ORDER OF THE LETTERS IS PART OF THE STATEMENT

$\triangle ABC \equiv \triangle A_1B_1C_1$  says  $A$  matches  $A_1$ ,  $B$  matches  $B_1$  and  $C$  matches  $C_1$ . Writing them in the wrong order claims something false.

The criterion

If two sides of one triangle and the angle between them are equal to two sides and the angle between them of another, the triangles are congruent

$$AB = A_1B_1, AC = A_1C_1, \angle A = \angle A_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

Three of the six pairs are enough — provided they are the **right** three.

| GIVEN                              | CONGRUENT?                |
|------------------------------------|---------------------------|
| two sides and the included angle   | yes — the first criterion |
| two sides and a non-included angle | not necessarily           |
| three angles                       | no — only similar         |

WHY THE ANGLE MUST BE INCLUDED

Two sides fixed at a hinge open and close: the third side changes with the angle. Fixing that angle fixes the triangle. With a non-included angle the hinge is not fixed and two shapes remain possible.

Writing a proof

**Problem.**  $AB = AD$  and  $\angle BAC = \angle DAC$ . Prove  $\triangle ABC \equiv \triangle ADC$ .

| STATEMENT                            | REASON                |
|--------------------------------------|-----------------------|
| $AB = AD$                            | given                 |
| $\angle BAC = \angle DAC$            | given                 |
| $AC = AC$                            | common side           |
| $\triangle ABC \equiv \triangle ADC$ | first criterion (SAS) |

THE COMMON SIDE IS THE MARK MOST OFTEN MISSED

Almost every proof of this kind needs it, and it is always available. Looking for a shared side or a shared angle is the first thing to do in any congruence question.

### Consequences

Once a congruence is proved, **all six** pairs of elements are equal. That is what makes the criteria useful: three facts are given, and three more follow free.

| PROVED                               | THEN ALSO                 |
|--------------------------------------|---------------------------|
| $\triangle ABC \equiv \triangle ADC$ | $BC = DC$                 |
| same                                 | $\angle B = \angle D$     |
| same                                 | $\angle BCA = \angle DCA$ |

The abbreviation used in proofs is *CPCT* — corresponding parts of congruent triangles.

MATCH THE ELEMENTS BY POSITION, NOT BY APPEARANCE

In  $\triangle ABC \equiv \triangle ADC$ ,  $BC$  corresponds to  $DC$  because  $B$  matches  $D$  and  $C$  matches  $C$ . Reading the correspondence off the letters is always safe; reading it off the drawing is not.

## 02 Worked examples

**EXAMPLE 1**  $AB = AD$  and  $\angle BAC = \angle DAC$ . Prove  $\triangle ABC \equiv \triangle ADC$ .

- 1  $AB = AD$  — given.
- 2  $\angle BAC = \angle DAC$  — given.  
The included angle.
- 3  $AC$  is common to both.  
The extra fact.
- 4 By the first criterion,  $\triangle ABC \equiv \triangle ADC$ .

ANSWER

Proved by SAS

**EXAMPLE 2** In the same figure, what else follows?

- 1 All six pairs of elements are equal.  
CPCT.
- 2  $BC = DC$
- 3  $\angle B = \angle D$
- 4  $\angle BCA = \angle DCA$

ANSWER

$BC = DC$ ,  $\angle B = \angle D$ ,  $\angle BCA = \angle DCA$



**EXAMPLE 3** Two triangles have sides 6 and 8 with an angle of  $40^\circ$ . Must they be congruent?

- ① If the  $40^\circ$  lies between the 6 and the 8: yes.  
SAS.
- ② If it lies elsewhere: not necessarily.
- ③ The position of the angle decides.
- ④ Only when it is the included angle.

**ANSWER**

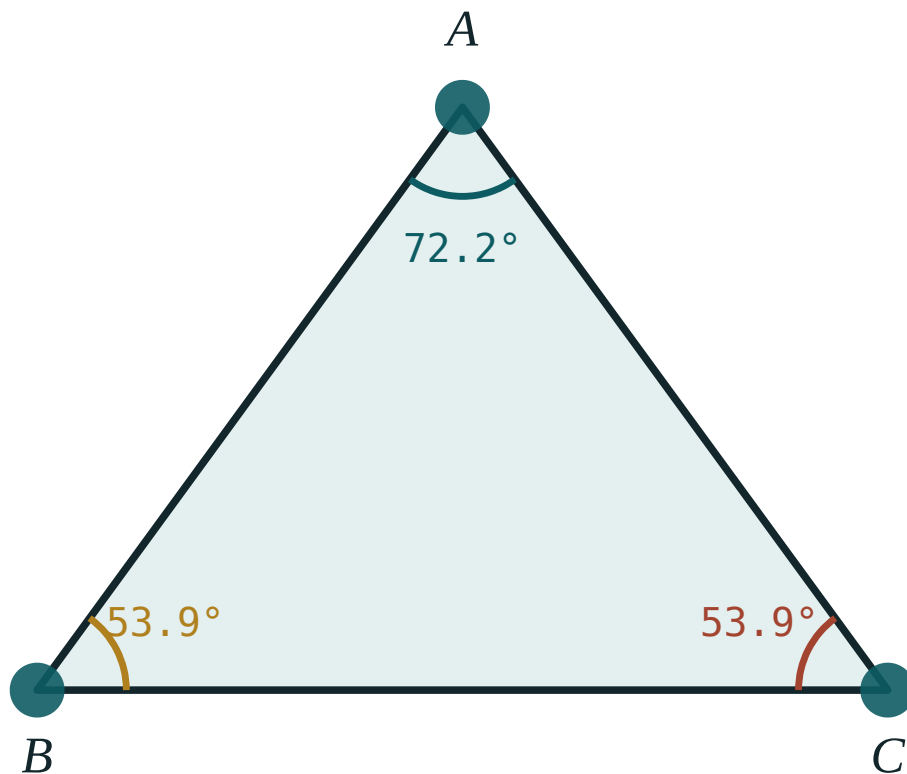
Only if the angle is included

**03 Interactive model**

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Hinge two rulers at a fixed angle and mark the third side; changing the angle changes it, and the criterion becomes a physical fact.

## Two sides and the angle between

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side  $BC$ largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                | O'ZBEKCHA         | РУССКИЙ                  |
|------------------------|-------------------|--------------------------|
| Congruent              | Teng              | Равный                   |
| Criterion              | Alomat            | Признак                  |
| Corresponding elements | Mos elementlar    | Соответственные элементы |
| Included angle         | Orasidagi burchak | Угол между сторонами     |
| Superposition          | Ustma-ust qo'yish | Наложение                |
| To follow              | Kelib chiqmoq     | Следовать                |
| Common side            | Umumiy tomon      | Общая сторона            |
| Proof                  | Isbot             | Доказательство           |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Congruent triangles have how many pairs of equal elements?

The first criterion needs:

Three equal angles give:

A shared side in a figure is:

irrelevant

a usable equality

a mistake

an angle

*CPCT* means:

a criterion

corresponding parts of congruent triangles

a construction

a theorem name

A non-included equal angle gives:

congruence

not necessarily congruence

similarity only

nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

**Easy · warm the idea up**

7 PROBLEMS

1. Congruent triangles have how many equal pairs?

ANSWER 6

2. The symbol for congruence

ANSWER  $\equiv$

3. The first criterion in letters

ANSWER SAS

4. A common side gives

ANSWER An equality for free

5. Three equal angles give

ANSWER Similarity only

6. In  $\triangle ABC \equiv \triangle DEF$ ,  $AB$  matches

ANSWER  $DE$

7. In the same,  $\angle C$  matches

ANSWER  $\angle F$

Medium · the standard question

7 PROBLEMS

1.  $AB = AD$ ,  $\angle BAC = \angle DAC$ : the third fact

ANSWER  $AC$  common

2. What follows about  $BC$ ?

ANSWER  $BC = DC$

3. What follows about  $\angle B$ ?

ANSWER  $\angle B = \angle D$

4. Sides 6, 8 with  $40^\circ$  included: congruent?

ANSWER Yes

5. Sides 6, 8 with  $40^\circ$  not included: congruent?

ANSWER Not necessarily

6. Two triangles with 5, 7 and  $60^\circ$  included

ANSWER Congruent

7. Are congruent triangles similar?

ANSWER Yes, with  $k = 1$

### Hard · stretch and reason

7 PROBLEMS

1. In a parallelogram, prove the two triangles cut by a diagonal are congruent

ANSWER SAS with the common diagonal

2. Prove that the diagonals of a parallelogram bisect each other

ANSWER Congruent triangles at the centre

3. An isosceles triangle: prove the bisector from the apex is a median

ANSWER SAS

4. Two triangles with  $AB = DE$ ,  $AC = DF$ ,  $\angle A = \angle D$ : name the criterion

ANSWER The first

5. Why is AAA not enough?

ANSWER Any enlargement has the same angles

6. In  $\triangle ABC \equiv \triangle DEF$  with  $AB = 5$ ,  $\angle B = 40^\circ$ : find  $DE$  and  $\angle E$

ANSWER 5 and  $40^\circ$

7. Prove that a kite has two pairs of congruent triangles

ANSWER SAS on each diagonal

## 07 Homework

### Homework — 5 tasks

Look for a common side or a common angle in every figure before starting.

- In a figure with  $AB = AC$  and  $\angle BAD = \angle CAD$ , prove  $\triangle ABD \equiv \triangle ACD$ .
- State what else follows from that congruence.
- Explain why the equal angle must lie between the two equal sides.
- Two triangles have sides 7 and 9 with an included angle of  $55^\circ$ . Are they congruent?
- Prove that a diagonal of a parallelogram divides it into two congruent triangles.



# Properties of isosceles triangles

Two equal sides force two equal angles — and one line does the work of three.

LESSON 49–50

2 × 40 MIN

MATEMATIKA 7, §21

S8 5.2

LESSON CLOCK

13'

23'

22'

18'

4'

|                       |        |
|-----------------------|--------|
| The base angles       | 13 min |
| The proof             | 23 min |
| The converse          | 22 min |
| One line, three roles | 18 min |
| Homework              | 4 min  |

LEARNING OBJECTIVES

- State and prove that the base angles of an isosceles triangle are equal.
- State and use the converse.
- Know that the bisector from the apex is also the median and the altitude.
- Solve numerical problems on isosceles and equilateral triangles.

TEXTBOOK

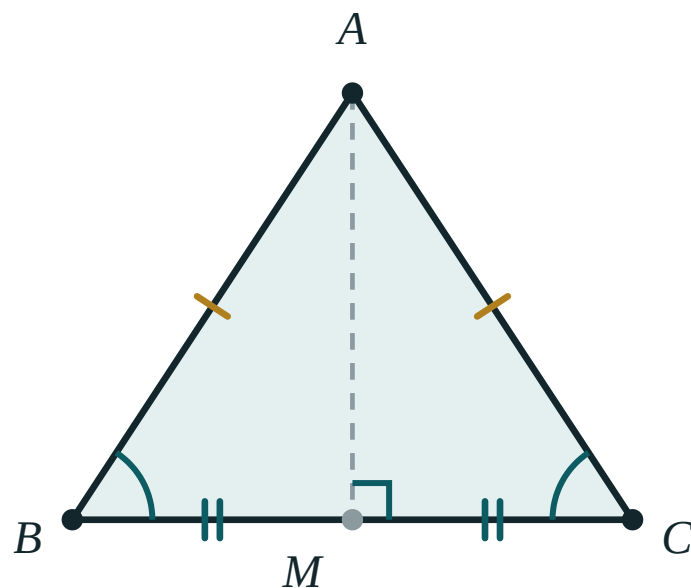
|           |                    |
|-----------|--------------------|
| National  | pp. 130–136        |
| Cambridge | Stage 8, pp. 50–53 |
| Workbook  | Exercise 5.2       |

01

## Explanation

### The base angles

In an isosceles triangle the angles at the base are equal



$$AB = AC \Rightarrow \angle B = \angle C$$

Two equal sides, two equal base angles — and one axis of symmetry.

|               |                                   |
|---------------|-----------------------------------|
| NAME          | IN $\triangle ABC$ WITH $AB = AC$ |
| lateral sides | $AB, AC$                          |
| base          | $BC$                              |
| base angles   | $\angle B, \angle C$              |
| apex angle    | $\angle A$                        |

Since the angles add to  $180^\circ$ :  $\angle B = \angle C = \frac{180^\circ - \angle A}{2}$ , and  $\angle A = 180^\circ - 2\angle B$ .

#### ONE ANGLE DETERMINES ALL THREE

Give any one angle of an isosceles triangle — apex or base — and the other two follow at once. That is what makes these questions quick.

### The proof

Draw the bisector  $AD$  of the apex angle.

| STATEMENT                            | REASON                  |
|--------------------------------------|-------------------------|
| $AB = AC$                            | given                   |
| $\angle BAD = \angle CAD$            | $AD$ bisects $\angle A$ |
| $AD = AD$                            | common                  |
| $\triangle ABD \equiv \triangle ACD$ | first criterion         |
| $\angle B = \angle C$                | CPCT                    |

#### THE AUXILIARY LINE IS THE WHOLE PROOF

Without  $AD$  there are no triangles to compare. Drawing the right extra line is the skill this proof teaches, and it recurs throughout the year.

## The converse

If two angles of a triangle are equal, the sides opposite them are equal

So a triangle is isosceles **if and only if** two of its angles are equal. Either fact may be used to prove the other.

| GIVEN                  | CONCLUDE                    |
|------------------------|-----------------------------|
| $AB = AC$              | $\angle B = \angle C$       |
| $\angle B = \angle C$  | $AB = AC$                   |
| all three sides equal  | all three angles $60^\circ$ |
| all three angles equal | all three sides equal       |

#### A CONVERSE MUST BE PROVED SEPARATELY

That a statement is true does not make its converse true. Here both directions happen to hold, but each needs its own proof — and in general they can differ.

## One line, three roles

In an isosceles triangle the bisector from the apex is also the median and the altitude to the base

It is also the axis of symmetry of the triangle. Four descriptions of one segment.

| THE SEGMENT $AD$        | BECAUSE                              |
|-------------------------|--------------------------------------|
| bisects $\angle A$      | by construction                      |
| $BD = DC$               | CPCT — it is a median                |
| $AD \perp BC$           | $\angle ADB = \angle ADC = 90^\circ$ |
| is the axis of symmetry | the two halves are congruent         |

ONLY FROM THE APEX

The bisector from a **base** angle is not a median and not an altitude. The property belongs to the apex alone.

02 Worked examples

EXAMPLE 1 An isosceles triangle has apex angle  $40^\circ$ . Find the base angles.

1

The three angles add to  $180^\circ$ .

2

$180^\circ - 40^\circ = 140^\circ$  for the two base angles.

3

They are equal.

4

$70^\circ$  each.

ANSWER

$70^\circ$  each

**EXAMPLE 2** An isosceles triangle has a base angle of  $72^\circ$ . Find the apex angle.

① Both base angles are  $72^\circ$ .

②  $72 + 72 = 144$

③  $180 - 144$

④  $= 36^\circ$

ANSWER

$36^\circ$

**EXAMPLE 3** In  $\triangle ABC$ ,  $\angle B = \angle C = 55^\circ$  and  $BC = 12$ . What can be said about the sides, and about the altitude from A?

① Equal angles  $\Rightarrow$  equal opposite sides:  $AB = AC$ .  
The converse.

② So the triangle is isosceles with base  $BC$ .

③ The altitude from A is also the median.

④ It meets  $BC$  at its midpoint, 6 from each end.

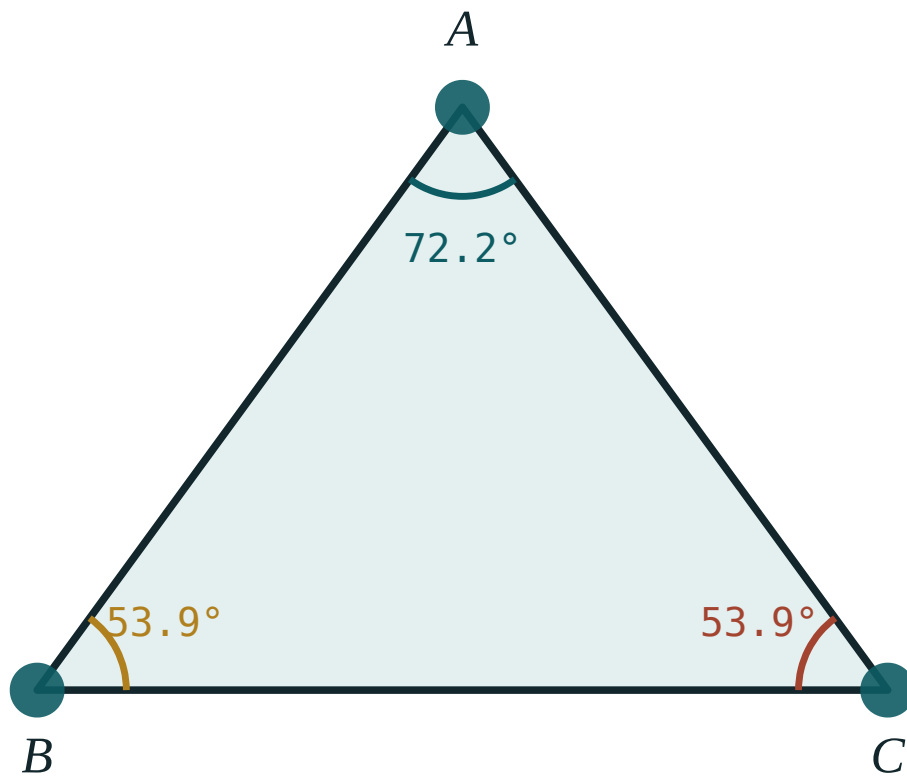
ANSWER

$AB = AC$ ; the altitude bisects  $BC$

### 03 Interactive model

Fold a paper isosceles triangle along the line from the apex; the two halves coincide, and all four properties of that line appear at once.

## Isosceles angles

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side  $BC$ largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA             | РУССКИЙ                    |
|--------------------|-----------------------|----------------------------|
| Isosceles triangle | Teng yonli uchburchak | Равнобедренный треугольник |
| Lateral side       | Yon tomon             | Боковая сторона            |
| Base               | Asos                  | Основание                  |
| Base angle         | Asosdagi burchak      | Угол при основании         |
| Apex angle         | Uchidagi burchak      | Угол при вершине           |
| Converse           | Teskari teorema       | Обратная теорема           |
| Axis of symmetry   | Simmetriya o'qi       | Ось симметрии              |
| Property           | Xossa                 | Свойство                   |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

In an isosceles triangle the equal angles are:

at the apex

at the base

all three

none

Apex  $40^\circ$  gives base angles:

$40^\circ$

$70^\circ$

$140^\circ$

$100^\circ$

Base angle  $72^\circ$  gives apex:

$36^\circ$

$72^\circ$

$108^\circ$

$144^\circ$



Two equal angles imply:

nothing

two equal sides

a right angle

equal areas

The bisector from the apex is also:

a median only

an altitude only

both

neither

An equilateral triangle has angles:

$45^\circ$

$60^\circ$

$90^\circ$

various

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Apex  $40^\circ$ : base angles

ANSWER  $70^\circ$

2. Apex  $90^\circ$ : base angles

ANSWER  $45^\circ$

3. Base angle  $72^\circ$ : apex

ANSWER  $36^\circ$

4. Base angle  $50^\circ$ : apex

ANSWER  $80^\circ$

5. Angles of an equilateral triangle

ANSWER  $60^\circ$

6. Two equal angles imply

ANSWER Two equal sides

7. The bisector from the apex is also

ANSWER A median and an altitude

Medium · the standard question

7 PROBLEMS

1. Apex  $100^\circ$ : base angles

ANSWER  $40^\circ$

2. Base angle  $35^\circ$ : apex

ANSWER  $110^\circ$

3. An isosceles triangle with a  $90^\circ$  apex: base angles

ANSWER  $45^\circ$

4. An isosceles triangle with a  $90^\circ$  base angle: possible?

ANSWER No

5.  $\angle B = \angle C = 55^\circ$ :  $\angle A$

ANSWER  $70^\circ$

6. Perimeter 26, base 10: the lateral sides

ANSWER 8 each

7. Perimeter 30, lateral side 11: the base

ANSWER 8

**Hard · stretch and reason**

7 PROBLEMS

1. An isosceles triangle with base 10 and lateral side 13: its height

ANSWER 12

2. Its area

ANSWER 60

3. An isosceles triangle in which the apex is twice a base angle

ANSWER  $90^\circ, 45^\circ, 45^\circ$

4. An isosceles triangle in which a base angle is twice the apex

ANSWER  $36^\circ, 72^\circ, 72^\circ$

5. An exterior angle at the base of an isosceles triangle is  $130^\circ$ : the apex

ANSWER  $80^\circ$

6. Why can an isosceles triangle not have two obtuse angles?

ANSWER The sum would exceed  $180^\circ$

7. A triangle with angles  $x, x, 4x$

ANSWER  $30^\circ, 30^\circ, 120^\circ$

## 07 Homework

### Homework — 5 tasks

Say which angle is the apex and which are the base angles before computing.

1. An isosceles triangle has apex angle  $64^\circ$ . Find the base angles.
2. An isosceles triangle has a base angle of  $43^\circ$ . Find the apex angle.
3. An isosceles triangle has perimeter 34 and base 12. Find the lateral sides.
4. Prove that the base angles of an isosceles triangle are equal.
5. An isosceles triangle has base 16 and lateral side 17. Find its height and area.

# The second criterion of congruence of triangles

A side and the two angles on it — the criterion that surveyors use.

LESSON 51–52

2 × 40 MIN

MATEMATIKA 7, §22

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

|                      |        |
|----------------------|--------|
| The criterion        | 13 min |
| Which side counts    | 23 min |
| In proofs            | 22 min |
| Measuring a distance | 18 min |
| Homework             | 4 min  |

LEARNING OBJECTIVES

- State and apply the ASA criterion.
- Recognise when the given side is between the two angles.
- Use the criterion in proofs involving parallel lines.
- Apply it to measuring an inaccessible distance.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 137–142        |
| Cambridge | Stage 8, pp. 48–56 |
| Workbook  | Exercise 5.1       |

01

## Explanation

### The criterion

If a side and the two angles on it in one triangle equal a side and the two angles on it in another, the triangles are congruent

$$BC = B_1C_1, \angle B = \angle B_1, \angle C = \angle C_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

The third angle is then equal too, since the angles of a triangle add to 180°.

TWO ANGLES AND ANY SIDE ARE ENOUGH

Because the third angle follows from the angle sum, a side and any two angles determine the triangle — whether the side lies between them or not. That is why the criterion is sometimes written ASA and sometimes AAS.

### Which side counts

| GIVEN                          | ENOUGH? | WHY                             |
|--------------------------------|---------|---------------------------------|
| $a, \angle B, \angle C$        | yes     | the side is between them        |
| $a, \angle A, \angle B$        | yes     | $\angle C$ follows from the sum |
| $\angle A, \angle B, \angle C$ | no      | no side fixes the size          |

The middle row is the one that surprises: the side need not be included, because the missing angle is never really missing.

### THREE ANGLES FIX THE SHAPE, NEVER THE SIZE

Any enlargement of a triangle has the same three angles. At least one length is always needed.

## In proofs

**Problem.**  $AB \parallel CD$  and  $AB = CD$ . Prove  $\triangle ABO \equiv \triangle DCO$ , where  $O$  is the intersection of  $AD$  and  $BC$ .

| STATEMENT                            | REASON                              |
|--------------------------------------|-------------------------------------|
| $AB = CD$                            | given                               |
| $\angle ABO = \angle DCO$            | alternate angles, $AB \parallel CD$ |
| $\angle BAO = \angle CDO$            | alternate angles                    |
| $\triangle ABO \equiv \triangle DCO$ | second criterion                    |

### PARALLEL LINES ARE A SUPPLY OF EQUAL ANGLES

Almost every use of the second criterion begins with a pair of parallels. Spotting them turns a hard figure into an easy proof.

## Measuring a distance

To measure the width of a river without crossing it:

1. mark  $A$  on one bank and a landmark  $B$  on the other;
2. walk along the bank to  $C$  and measure  $AC$ ;
3. measure  $\angle BAC$  and  $\angle BCA$ ;

4. construct a triangle with the same side and two angles on level ground, and measure its third side.

By the second criterion the constructed triangle is congruent to the real one, so the measured side **is** the width of the river.

THIS IS WHAT SURVEYING WAS, BEFORE TRIGONOMETRY

A tape and a protractor, a criterion of congruence, and a drawing on level ground. The method is ancient, exact, and still taught for exactly that reason.

## 02 Worked examples

**EXAMPLE 1** Two triangles have a side of 7 with angles  $50^\circ$  and  $60^\circ$  on it. Are they congruent?

- ① A side and the two angles on it.
- ② The second criterion applies.
- ③ The third angle is  $70^\circ$  in both.
- ④ Congruent.

ANSWER

Yes, by ASA

**EXAMPLE 2** Two triangles have a side of 7, an angle of  $50^\circ$  on it and an angle of  $70^\circ$  opposite it. Congruent?

- 1 The third angle is  $180 - 50 - 70 = 60^\circ$ .
- 2 So all three angles are known in both.
- 3 With the side of 7 in the same position, ASA applies.
- 4 Congruent.  
The side need not be included.

ANSWER

Yes

**EXAMPLE 3**  $AB \parallel CD$  and  $AB = CD$ . Prove that  $AD$  and  $BC$  bisect each other.

- 1 Alternate angles give  $\angle ABO = \angle DCO$  and  $\angle BAO = \angle CDO$ .
- 2 With  $AB = CD$ , the second criterion gives  $\triangle ABO \equiv \triangle DCO$ .
- 3 CPCT:  $AO = DO$  and  $BO = CO$ .
- 4 So each diagonal is bisected.

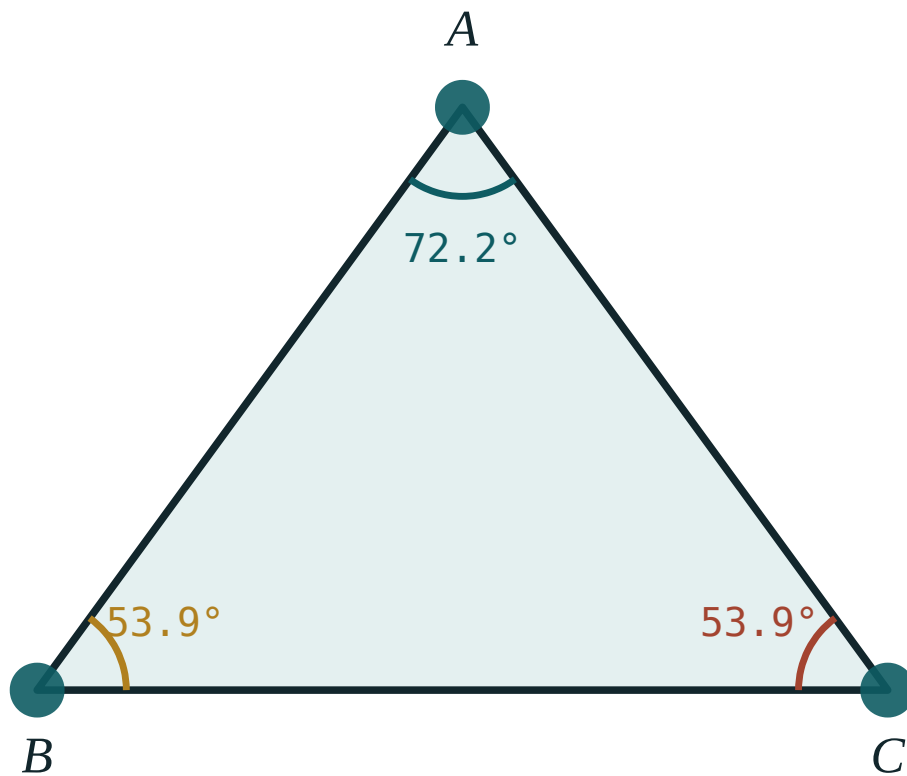
ANSWER

Proved

### 03 Interactive model

Measure the width of the corridor from inside the classroom using a tape and a protractor; the construction on paper gives an answer that can then be checked directly.



**A side and two angles** $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side  $BC$ largest angle  $\angle A$ **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH               | O'ZBEKCHA                             | РУССКИЙ                     |
|-----------------------|---------------------------------------|-----------------------------|
| Second criterion      | <b>Ikkinchi alomat</b>                | <b>Второй признак</b>       |
| Two angles and a side | <b>Ikki burchak va tomon</b>          | <b>Два угла и сторона</b>   |
| Adjacent side         | <b>Yondosh tomon</b>                  | <b>Прилежащая сторона</b>   |
| Angle sum             | <b>Burchaklar yig'indisi</b>          | <b>Сумма углов</b>          |
| Surveying             | <b>Yer o'lchash</b>                   | <b>Съёмка местности</b>     |
| Inaccessible          | <b>Yetib bo'lmaydigan</b>             | <b>Недоступный</b>          |
| Alternate angles      | <b>Ichki almashinuvchi burchaklar</b> | <b>Накрест лежащие углы</b> |
| Proof                 | <b>Isbot</b>                          | <b>Доказательство</b>       |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

The second criterion needs:

three sides

a side and two angles

two sides and an angle

three angles

The third angle:

must be given

follows from the sum

is irrelevant

is always  $90^\circ$

Must the side lie between the angles?

yes

no

only sometimes

it depends on the size

Three angles alone give:

congruence

similarity

nothing

equal areas

Parallel lines supply:

equal sides

equal angles

right angles

nothing

Surveying a river width uses:

the first criterion

the second criterion

the triangle inequality

Pythagoras

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The second criterion in letters

ANSWER ASA

2. A side of 7 with  $50^\circ$  and  $60^\circ$  on it: third angle

ANSWER  $70^\circ$

3. Two triangles with that data

ANSWER Congruent

4. Three equal angles give

ANSWER Similarity only

5. Alternate angles come from

ANSWER Parallel lines

6. Angles  $50^\circ$ ,  $70^\circ$ : the third

ANSWER  $60^\circ$

7. Is a side always needed?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. A side of 7,  $50^\circ$  on it and  $70^\circ$  opposite: congruent?

ANSWER Yes

2.  $AB \parallel CD$ ,  $AB = CD$ : which criterion?

ANSWER The second

3. What follows about  $AO$  and  $DO$ ?

ANSWER They are equal

4. Two triangles with  $\angle A = \angle D$ ,  $\angle B = \angle E$ ,  $AB = DE$

ANSWER Congruent

5. Two triangles with  $\angle A = \angle D$ ,  $\angle B = \angle E$  only

ANSWER Similar only

6. Angles  $40^\circ$ ,  $40^\circ$  with the included side 9

ANSWER Congruent to any such triangle

7. The third angle there

ANSWER  $100^\circ$

**Hard · stretch and reason**

7 PROBLEMS

1. Prove that the diagonals of a parallelogram bisect each other

ANSWER ASA with alternate angles

2. Prove that a point on the bisector of an angle is equidistant from the arms

ANSWER ASA, then CPCT

3. Prove that the base angles of an isosceles triangle are equal, using ASA

ANSWER Compare the triangle with itself, reversed

4. A triangle with  $\angle A = 3x$ ,  $\angle B = 2x$ ,  $\angle C = x$

ANSWER  $90^\circ$ ,  $60^\circ$ ,  $30^\circ$

5. Why is ASA enough but SSA not?

ANSWER The third angle is forced; the third side is not

6. Two triangles share a side and have two pairs of equal angles

ANSWER Congruent

7. How many measurements are needed to fix a triangle?

ANSWER Three, including at least one length

## 07 Homework

### Homework — 5 tasks

Mark the equal angles and the equal side on the figure before writing the proof.

1. Two triangles have a side of 9 with angles  $45^\circ$  and  $65^\circ$  on it. Are they congruent?
2. Find the third angle in that triangle.
3. Prove that the diagonals of a parallelogram bisect each other.
4. Explain why three angles alone do not prove congruence.
5. Describe how to measure the width of a river using the second criterion.

# The third criterion of congruence of triangles

Three sides and nothing else — the criterion that makes a triangle rigid.

LESSON 53–54

2 × 40 MIN

MATEMATIKA 7, §23

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

|                      |        |
|----------------------|--------|
| The criterion        | 13 min |
| Rigidity             | 23 min |
| Using it             | 22 min |
| Choosing a criterion | 18 min |
| Homework             | 4 min  |

LEARNING OBJECTIVES

- State and apply the SSS criterion.
- Use it to prove properties of the rhombus and the kite.
- Explain why a triangle is rigid and a quadrilateral is not.
- Choose between the three criteria from the given data.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 143–148        |
| Cambridge | Stage 8, pp. 48–56 |
| Workbook  | Exercise 5.1       |

01

## Explanation

### The criterion

If the three sides of one triangle are equal to the three sides of another, the triangles are congruent

$$AB = A_1B_1, BC = B_1C_1, CA = C_1A_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

No angle is mentioned at all, and yet all three angles turn out equal. Three lengths determine a triangle completely.

THIS IS THE CRITERION A COMPUTER WOULD USE

It needs no reasoning about the figure — only three comparisons. Every question that supplies six lengths and no angles is answered by it.

### Rigidity

A framework of three rods joined at their ends cannot change shape; one of four rods can.



| FRAMEWORK                       | RIGID? | WHY                                      |
|---------------------------------|--------|------------------------------------------|
| triangle                        | yes    | three sides fix it — the third criterion |
| quadrilateral                   | no     | it can be pushed into a parallelogram    |
| quadrilateral with one diagonal | yes    | it becomes two triangles                 |

This is why bridges, roof trusses and pylons are built from triangles, and why a garden gate is braced with a diagonal.

#### THE THIRD CRITERION IS AN ENGINEERING FACT

Every triangulated structure in the world depends on it. The theorem and the brace on a gate are the same statement.

## Using it

**Problem.** In a rhombus  $ABCD$ , prove that the diagonal  $AC$  bisects the angles at  $A$  and  $C$ .

| STATEMENT                            | REASON                         |
|--------------------------------------|--------------------------------|
| $AB = AD$                            | a rhombus has four equal sides |
| $CB = CD$                            | the same                       |
| $AC = AC$                            | common                         |
| $\triangle ABC \equiv \triangle ADC$ | third criterion                |
| $\angle BAC = \angle DAC$            | CPCT                           |

#### THE COMMON SIDE AGAIN

As in the first criterion, the shared side supplies the third equality without being given. Look for it first in every figure.

## Choosing a criterion

| GIVEN                                | CRITERION              |
|--------------------------------------|------------------------|
| two sides and the angle between them | the first (SAS)        |
| a side and two angles                | the second (ASA)       |
| three sides                          | the third (SSS)        |
| two sides and a non-included angle   | none — not enough      |
| three angles                         | none — similarity only |

NAME THE CRITERION IN EVERY PROOF

A proof that reaches the right conclusion without naming which criterion was used is incomplete, and is marked as such in both the national and the Cambridge papers.

02

## Worked examples

---

EXAMPLE 1 Two triangles have sides 5, 6, 7 and 7, 5, 6. Are they congruent?

1

Sort both lists: 5, 6, 7 and 5, 6, 7.

2

All three pairs are equal.

3

By the third criterion, congruent.

4

The order in which they were written does not matter.

ANSWER

Yes, by SSS

**EXAMPLE 2** In rhombus  $ABCD$ , prove  $\triangle ABC \equiv \triangle ADC$ .

- ①  $AB = AD$  — all sides of a rhombus are equal.
- ②  $CB = CD$  — the same.
- ③  $AC$  is common.
- ④ Third criterion.

ANSWER

Proved by SSS

**EXAMPLE 3** Why is a four-rod framework not rigid, and how is it braced?

- ① Four sides do not determine the angles.
- ② The frame can be pushed out of square.
- ③ Adding one diagonal makes two triangles.
- ④ Each is rigid by SSS, so the whole frame is.

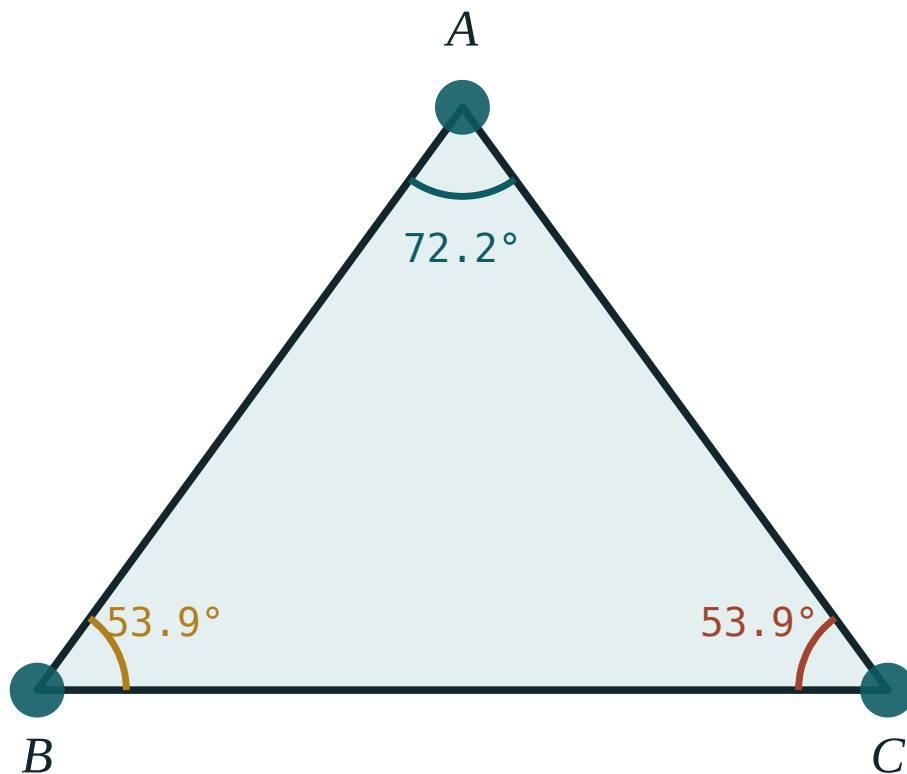
ANSWER

Add a diagonal

### 03 Interactive model

Pin three strips of card into a triangle and four into a quadrilateral; the class pushes both, and the third criterion becomes visible in ten seconds.

## Three sides fix the triangle

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side **BC**largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH         | O'ZBEKCHA       | РУССКИЙ        |
|-----------------|-----------------|----------------|
| Third criterion | Uchinchi alomat | Третий признак |
| Three sides     | Uchta tomon     | Три стороны    |
| Rigid           | Qattiq          | Жёсткий        |
| Rhombus         | Romb            | Ромб           |
| Kite            | Deltoid         | Дельтоид       |
| Diagonal        | Diagonal        | Диагональ      |
| Framework       | Karkas          | Каркас         |
| To choose       | Tanlash         | Выбрать        |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

The third criterion needs:

three angles

three sides

two sides and an angle

a side and two angles

A triangular framework is:

flexible

rigid

unstable

impossible

A quadrilateral framework is made rigid by:

a fifth rod along a side

a diagonal

nothing

a shorter rod

Sides 5, 6, 7 and 7, 5, 6 give:

congruence

similarity only

nothing

an error

Two sides and a non-included angle give:

congruence

not necessarily

similarity

nothing at all

A complete proof must:

give the answer

name the criterion

include measurements

be short

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The third criterion in letters

ANSWER SSS

2. Sides 5, 6, 7 and 5, 6, 7

ANSWER Congruent

3. Sides 5, 6, 7 and 5, 6, 8

ANSWER Not congruent

4. Is a triangle rigid?

ANSWER Yes

5. Is a quadrilateral rigid?

ANSWER No

6. How is it braced?

ANSWER With a diagonal

7. A shared side gives

ANSWER An equality for free

Medium · the standard question

7 PROBLEMS



1. Sides 5, 6, 7 and 7, 5, 6

ANSWER Congruent

2. Which criterion for six lengths?

ANSWER The third

3. Which for a side and two angles?

ANSWER The second

4. Which for two sides and the included angle?

ANSWER The first

5. Which for three angles?

ANSWER None

6. In a rhombus,  $\triangle ABC$  and  $\triangle ADC$

ANSWER Congruent by SSS

7. What follows about  $\angle BAC$  and  $\angle DAC$ ?

ANSWER They are equal

Hard · stretch and reason

7 PROBLEMS

1. Prove that a diagonal of a rhombus bisects its angles

ANSWER SSS, then CPCT

2. Prove that the diagonals of a rhombus are perpendicular

ANSWER Congruent triangles at the centre

3. Prove that a kite has one diagonal as an axis of symmetry

ANSWER SSS on the two triangles

4. Why is a bridge built of triangles?

ANSWER Only triangles are rigid

5. Two triangles with sides  $a, b, c$  and  $c, b, a$

ANSWER Congruent

6. Can two triangles have five equal elements and not be congruent?

ANSWER Yes — similar triangles can

7. How many rods brace a quadrilateral?

ANSWER One diagonal

## 07 Homework

### Homework — 5 tasks

Sort both lists of sides before comparing, and name the criterion in every proof.

- Are triangles with sides  $8, 9, 10$  and  $10, 8, 9$  congruent?
- Prove that a diagonal of a rhombus bisects two of its angles.
- Explain why a triangle is rigid and a quadrilateral is not.
- State which criterion applies to each of the five cases in the table.
- Prove that the diagonals of a rhombus are perpendicular.

# Practical exercises

## — constructions

### with compasses

### and a straightedge

A Cambridge insert: the four constructions every later proof depends on.

LESSON 55

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 5.3

LESSON CLOCK

8'

14'

12'

6'

|                        |        |
|------------------------|--------|
| The tools              | 8 min  |
| The four constructions | 14 min |
| Building a triangle    | 12 min |
| Homework               | 6 min  |

LEARNING OBJECTIVES

- Copy a segment and an angle with compasses.
- Construct the perpendicular bisector of a segment.
- Construct the bisector of an angle.
- Construct a triangle from three given sides.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 149–151        |
| Cambridge | Stage 8, pp. 53–56 |
| Workbook  | Exercise 5.3       |

01

## Explanation

### The tools

A **construction** uses only two tools: compasses to draw circles and to carry lengths, and a straightedge to draw lines. The straightedge is **not** a ruler — no measuring is allowed.

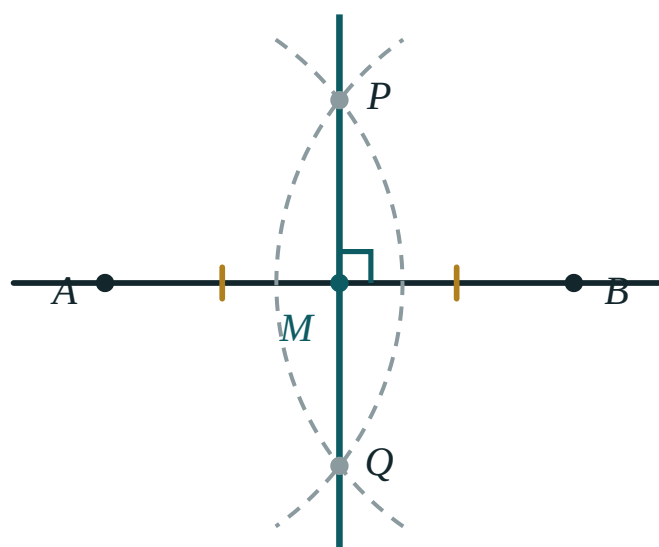
| TOOL         | MAY BE USED TO              | MAY NOT                |
|--------------|-----------------------------|------------------------|
| compasses    | draw an arc, carry a length | measure in centimetres |
| straightedge | join two points             | measure                |

LEAVE EVERY ARC VISIBLE

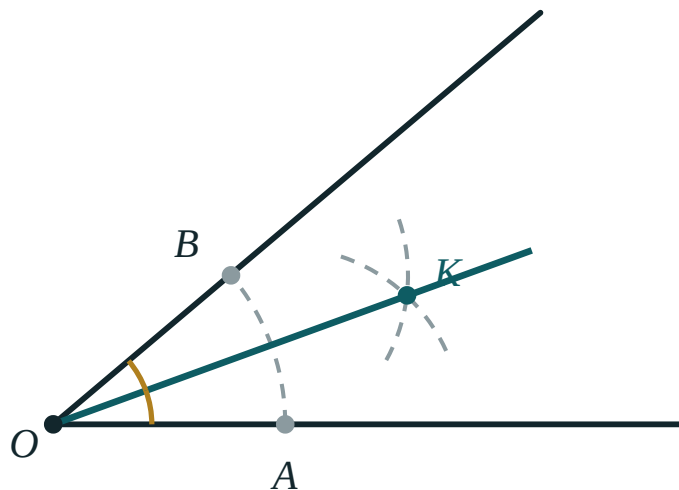
The arcs are the working. A finished construction with the arcs rubbed out cannot be marked, and scores less than an untidy one that shows them.

### The four constructions

| CONSTRUCTION                   | METHOD                                                                            |
|--------------------------------|-----------------------------------------------------------------------------------|
| copy a segment                 | open the compasses to it and step it off on a ray                                 |
| copy an angle                  | one arc across both arms of the original and of the copy, then transfer the chord |
| perpendicular bisector of $AB$ | equal arcs from $A$ and from $B$ , both sides; join the crossings                 |
| bisector of an angle           | one arc across both arms, then equal arcs from the two points; join to the vertex |



*Equal arcs from both ends give the perpendicular bisector.*



*One arc across the arms, then two equal arcs, gives the bisector.*

#### ALL FOUR ARE THE SAME IDEA

Two points equidistant from the same pair of points determine a line of symmetry. Every one of the four constructions is that statement applied somewhere.

## Building a triangle

Given three sides  $a$ ,  $b$ ,  $c$ :

1. draw a segment  $BC = a$ ;
2. from  $B$ , an arc of radius  $c$ ;
3. from  $C$ , an arc of radius  $b$ ;
4. the crossing is  $A$ ; join  $AB$  and  $AC$ .

By the third criterion, every triangle built this way from the same three lengths is congruent to every other — so the construction gives **the** triangle, not **a** triangle.

#### THE ARCS MUST ACTUALLY CROSS

If the two short sides together are less than the long one, the arcs miss and no triangle exists. The construction fails exactly when the triangle inequality fails.

## 02 Worked examples

**EXAMPLE 1** Describe the construction of the perpendicular bisector of a segment  $AB$ .

- 1 Open the compasses to more than half of  $AB$ .
- 2 Draw arcs from  $A$  above and below the segment.
- 3 Draw the same arcs from  $B$ .  
Same radius.
- 4 Join the two crossings — that line is the perpendicular bisector.

ANSWER

Equal arcs from both ends, joined

**EXAMPLE 2** Construct a triangle with sides  $4\text{ cm}$ ,  $5\text{ cm}$  and  $6\text{ cm}$ .

- 1 Draw  $BC = 6\text{ cm}$ .  
The longest side is easiest.
- 2 Arc of  $4\text{ cm}$  from  $B$ .
- 3 Arc of  $5\text{ cm}$  from  $C$ .
- 4 Join the crossing to both ends.  
The arcs cross since  $4 + 5 > 6$ .

ANSWER

A unique triangle

**EXAMPLE 3** Why does no triangle exist with sides 3, 4, 9?

- ① Draw the 9 *cm* side.
- ② Arcs of 3 and 4 from its ends.
- ③  $3 + 4 = 7 < 9$  — the arcs fall short.
- ④ They never meet, so there is no third vertex.

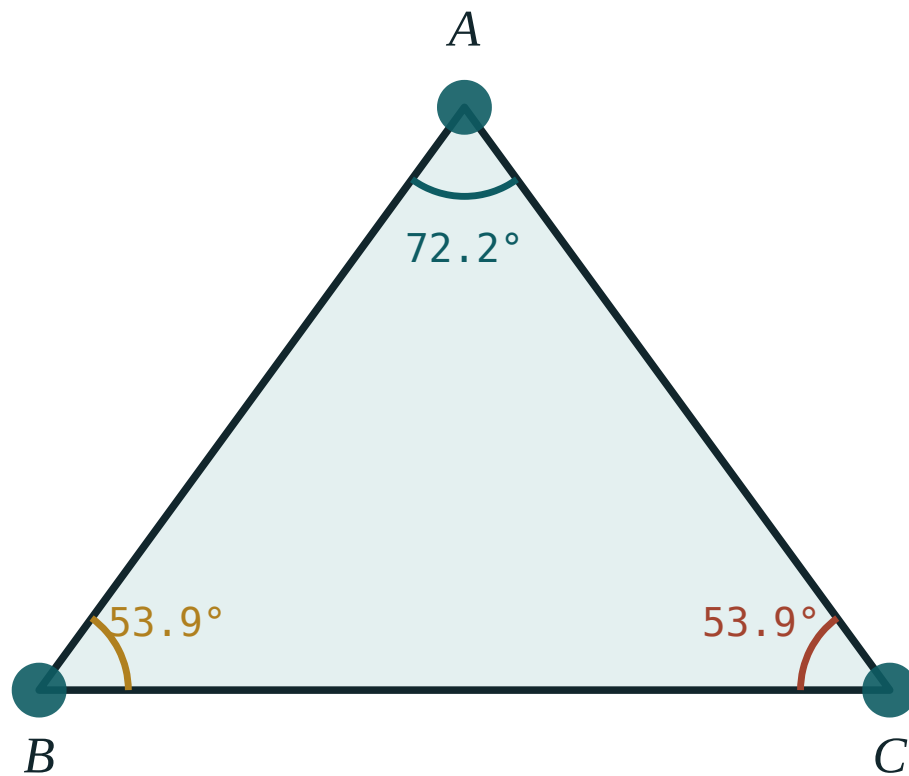
**ANSWER**

The arcs do not cross

**03** Interactive model

Have the class construct the same triangle from the same three lengths and then lay the sheets on top of one another; every triangle coincides, and the third criterion is confirmed.

## Building from three sides

 $\angle A$   $72.2^\circ$  $\angle B$   $53.9^\circ$  $\angle C$   $53.9^\circ$ sum  $180^\circ$ longest side **BC**largest angle  $\angle A$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.



| ENGLISH                | O'ZBEKCHA            | РУССКИЙ                  |
|------------------------|----------------------|--------------------------|
| Construction           | Yasash               | Построение               |
| Compasses              | Sirkul               | Циркуль                  |
| Straightedge           | Chizg'ich            | Линейка                  |
| Arc                    | Yoy                  | Дуга                     |
| To copy                | Ko'chirish           | Перенести                |
| Perpendicular bisector | O'rta perpendikulyar | Серединный перпендикуляр |
| Bisector               | Bissektrisa          | Биссектриса              |
| Equidistant            | Teng uzoqlikda       | Равноудалённый           |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A construction uses:

a ruler and a protractor

compasses and a straightedge

a calculator

a set square only

Construction arcs should be:

rubbed out

left visible

drawn in pen

omitted

The perpendicular bisector is found by:

measuring the midpoint

equal arcs from both ends

a protractor

folding only

A triangle from three sides is:

not always possible

always possible

possible in two ways

never unique

Sides 3, 4, 9:

one triangle

two triangles

no triangle

infinitely many

All triangles built from the same three sides are:

similar only

congruent

different

right-angled

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

## 1. Tools allowed

ANSWER Compasses and a straightedge

## 2. May the straightedge measure?

ANSWER No

## 3. Should arcs be left?

ANSWER Yes

## 4. Sides 4, 5, 6: constructible?

ANSWER Yes

## 5. Sides 3, 4, 9: constructible?

ANSWER No

## 6. Sides 5, 5, 5: what triangle?

ANSWER Equilateral

## 7. Sides 3, 4, 5: what triangle?

ANSWER Right-angled

Medium · the standard question

7 PROBLEMS

1. Which construction gives the midpoint?

ANSWER The perpendicular bisector

2. Which gives two equal angles?

ANSWER The angle bisector

3. How many arcs for a perpendicular bisector?

ANSWER Four — two from each end

4. Construct  $60^\circ$ : method

ANSWER An equilateral triangle

5. Construct  $30^\circ$ : method

ANSWER Bisect  $60^\circ$

6. Construct  $90^\circ$ : method

ANSWER A perpendicular

7. Construct  $45^\circ$ : method

ANSWER Bisect  $90^\circ$

### Hard · stretch and reason

7 PROBLEMS

1. Construct  $15^\circ$

ANSWER Bisect  $30^\circ$

2. Construct  $75^\circ$

ANSWER  $60^\circ + 15^\circ$

3. Construct an isosceles triangle with base 5 and lateral side 7

ANSWER Two arcs of 7

4. Why must the compasses open beyond half of  $AB$ ?

ANSWER Otherwise the arcs do not meet

5. Construct the centre of a given circle

ANSWER Two chords and their perpendicular bisectors

6. Construct a triangle from two sides and the included angle

ANSWER Copy the angle, step off the sides

7. Which construction cannot be done with these tools?

ANSWER Trisecting a general angle

## 07 Homework

### Homework — 5 tasks

Leave every arc on the page; the arcs are the answer.

1. Construct the perpendicular bisector of a segment of  $8\text{ cm}$ .
2. Construct the bisector of an angle of  $80^\circ$ .
3. Construct a triangle with sides  $5\text{ cm}$ ,  $6\text{ cm}$  and  $7\text{ cm}$ .
4. Construct an angle of  $30^\circ$  using only compasses and a straightedge.
5. Explain why no triangle can be constructed with sides 2, 3, 8.

# Control work 4, and work on the mistakes

The congruence of triangles in one paper, and the geometry chapter closed.

LESSON 56–57

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 4

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

|                       |        |
|-----------------------|--------|
| Instructions          | 3 min  |
| The paper             | 40 min |
| Answers               | 12 min |
| Diagnosis and rewrite | 20 min |
| The map               | 5 min  |

LEARNING OBJECTIVES

- Choose the right criterion from the given data.
- Write a proof in statement-and-reason form under time.
- Use the properties of isosceles triangles and the triangle inequality.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 118–151        |
| Cambridge | Stage 8, pp. 48–56 |
| Workbook  | Control paper M4   |

01

## Explanation

### The paper — 25 marks, 40 minutes

| Q | TASK                                                                                        | MARKS | FROM   |
|---|---------------------------------------------------------------------------------------------|-------|--------|
| 1 | Can a triangle have sides 5, 8, 14? And 6, 9, 13?                                           | 4     | L46    |
| 2 | An isosceles triangle has apex 44°: find the base angles                                    | 4     | L49–50 |
| 3 | $AB = AD$ , $\angle BAC = \angle DAC$ : prove $\triangle ABC \equiv \triangle ADC$          | 6     | L47–48 |
| 4 | Name the criterion for: three sides; a side and two angles; two sides and the angle between | 3     | L47–54 |
| 5 | In a rhombus, prove that a diagonal bisects two angles                                      | 5     | L53–54 |
| 6 | Describe the construction of an angle bisector                                              | 3     | L55    |

## WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for checking the longest side only; Q3 two for the common side and the named criterion; Q5 two for stating that all four sides of a rhombus are equal; Q6 one for “leave the arcs”.

## Naming the slip

| SLIP                          | WHAT IT LOOKS LIKE             | THE FIX                         |
|-------------------------------|--------------------------------|---------------------------------|
| only two sides checked        | $5 + 14 > 8$ ✓                 | check the longest: $5 + 8 < 14$ |
| apex used as a base angle     | $44^\circ, 44^\circ, 92^\circ$ | $44^\circ, 68^\circ, 68^\circ$  |
| common side omitted           | two facts only                 | $AC = AC$                       |
| criterion not named           | “so they are congruent”        | “by the first criterion”        |
| rhombus sides assumed unequal | “given $AB = AD$ ”             | state that all four are equal   |
| reasons missing               | a list of true statements      | a reason beside each            |
| arcs rubbed out               | a clean bisector               | leave the construction visible  |

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

## Chapter IV as one map

| BLOCK                   | THE SENTENCE                                           |
|-------------------------|--------------------------------------------------------|
| triangles               | classified by sides and by angles, independently       |
| the triangle inequality | the longest side is less than the sum of the other two |
| the first criterion     | two sides and the angle between them                   |
| isosceles triangles     | equal sides $\iff$ equal base angles                   |
| the second criterion    | a side and two angles                                  |
| the third criterion     | three sides — and rigidity                             |
| constructions           | compasses and a straightedge, arcs left visible        |



## LOOKING FORWARD

The rest of Quarter II is algebra: expressions, powers, monomials and polynomials. Geometry returns in Quarter III with parallel lines and the angle sum of a triangle.

## 02 Worked examples

**EXAMPLE 1** Model answer, Q1: sides 5, 8, 14 and 6, 9, 13.

- ① Longest of the first is 14.
- ②  $5 + 8 = 13 < 14$  — no triangle.
- ③ Longest of the second is 13.
- ④  $6 + 9 = 15 > 13$  — yes.

ANSWER

No; yes

**EXAMPLE 2** Model answer, Q3:  $AB = AD$ ,  $\angle BAC = \angle DAC$ .

- ①  $AB = AD$  — given.
- ②  $\angle BAC = \angle DAC$  — given, and it is the included angle.
- ③  $AC = AC$  — common side.  
The mark most often missed.
- ④  $\triangle ABC \equiv \triangle ADC$  by the first criterion.

ANSWER

Proved by SAS

**EXAMPLE 3** Model answer, Q5: a diagonal of a rhombus bisects two angles.

- ① All four sides of a rhombus are equal:  $AB = AD$  and  $CB = CD$ .
- ②  $AC$  is common.
- ③  $\triangle ABC \equiv \triangle ADC$  by the third criterion.
- ④ CPCT gives  $\angle BAC = \angle DAC$  and  $\angle BCA = \angle DCA$ .

**ANSWER**

Proved by SSS

**03 Interactive model**

Return Q3 with the reasons column blank and have the class fill it in; they discover that the reasons carry most of the marks.

**Chapter IV in twelve questions**

A triangle with two equal sides is:

scalene

isosceles

equilateral only

right

Can 2, 3, 6 be a triangle?

yes

no

sometimes

only if right

Congruent triangles have how many equal pairs?

3

4

6

2

The first criterion needs:

three sides

two sides and the included angle

a side and two angles

three angles

Base angles of an isosceles triangle are:

unequal

equal

right

obtuse

Apex  $40^\circ$  gives base angles:

$40^\circ$

$70^\circ$

$100^\circ$

$140^\circ$

The second criterion needs:

three sides

a side and two angles

three angles

two sides

The third criterion needs:

three sides

three angles

two sides

a side and an angle

A triangular framework is:

flexible

rigid

unstable

impossible

A common side in a figure gives:

nothing

an equality

an angle

a contradiction

Three equal angles give:

congruence

similarity only

nothing

rigidity

Construction arcs should be:

rubbed out

left visible

in pen

omitted

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH             | O'ZBEKCHA              | РУССКИЙ                  |
|---------------------|------------------------|--------------------------|
| Control work        | Nazorat ishi           | Контрольная работа       |
| Congruence          | Tenglik                | Равенство                |
| Criterion           | Alomat                 | Признак                  |
| Isosceles           | Teng yonli             | Равнобедренный           |
| Triangle inequality | Uchburchak tengsizligi | Неравенство треугольника |
| Common side         | Umumiy tomon           | Общая сторона            |
| Reason              | Asos                   | Обоснование              |
| Diagnosis           | Tashxis                | Диагностика              |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

For Q1 it is enough to check:

all three sums

the longest side

the shortest side

the perimeter

Q2 needs the angles to add to:

$90^\circ$

$180^\circ$

$270^\circ$

$360^\circ$

The missing fact in Q3 is:

an angle

the common side

a third side

nothing

Q5 uses:

SAS

ASA

SSS

none

Q6 must mention:

a protractor

leaving the arcs

measuring

a ruler

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1. Can 5, 8, 14 be a triangle?

ANSWER No

2. Can 6, 9, 13 be a triangle?

ANSWER Yes

3. Apex  $44^\circ$ : base angles

ANSWER  $68^\circ$

4. Three sides: which criterion?

ANSWER The third

5. A side and two angles: which?

ANSWER The second

6. Two sides and the included angle: which?

ANSWER The first

7. Three angles: which?

ANSWER None

Medium · the standard question

7 PROBLEMS

1.  $AB = AD$ ,  $\angle BAC = \angle DAC$ : the third fact

ANSWER  $AC$  common

2. That proof uses

ANSWER The first criterion

3. A rhombus diagonal proof uses

ANSWER The third criterion

4. Base angle  $68^\circ$ : apex

ANSWER  $44^\circ$

5. An isosceles triangle with perimeter 28 and base 8

ANSWER Sides 10, 10, 8

6. Sides 7, 7, 7: angles

ANSWER  $60^\circ$  each

7. Which framework is rigid?

ANSWER The triangle

### Hard · stretch and reason

7 PROBLEMS

1. Range of  $c$  with  $a = 6$ ,  $b = 11$

ANSWER  $5 < c < 17$

2. Integer values of  $c$  there

ANSWER 6 to 16

3. An isosceles triangle with an exterior base angle of  $115^\circ$ : the apex

ANSWER  $50^\circ$

4. Prove the base angles equal, using the bisector from the apex

ANSWER SAS then CPCT

5. Prove the diagonals of a rhombus are perpendicular

ANSWER SSS then equal adjacent angles

6. Two triangles with  $AB = DE$ ,  $\angle A = \angle D$ ,  $\angle B = \angle E$

ANSWER Congruent by ASA

7. Two triangles with  $AB = DE$ ,  $BC = EF$ ,  $\angle A = \angle D$

ANSWER Not necessarily congruent

## 07 Homework

### Homework — 5 tasks

Rewrite in full every proof that lost a mark, with a reason beside every statement.

- Can a triangle have sides 4, 7, 12? And 5, 7, 11?
- An isosceles triangle has apex angle  $52^\circ$ . Find the base angles.
- Prove that a diagonal of a rhombus bisects two of its angles.
- Name the criterion that applies to each of the three standard sets of data.
- Describe the construction of the bisector of an angle.

# Algebraic expressions

Letters standing for numbers — and the vocabulary the rest of the year needs.

LESSON 58

1 × 40 MIN

MATEMATIKA 7, §24

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

|                            |        |
|----------------------------|--------|
| Words into symbols         | 10 min |
| The parts of an expression | 12 min |
| Evaluating                 | 12 min |
| Homework                   | 6 min  |

LEARNING OBJECTIVES

- Write a worded statement as an algebraic expression.
- Name the terms, factors and coefficients of an expression.
- Evaluate an expression for given values of the letters.
- Distinguish an expression from an equation.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 152–155        |
| Cambridge | Stage 8, pp. 12–18 |
| Workbook  | Exercise 2.1       |

01

## Explanation

### Words into symbols

| WORDS                          | EXPRESSION    |
|--------------------------------|---------------|
| five more than $x$             | $x + 5$       |
| five less than $x$             | $x - 5$       |
| five times $x$                 | $5x$          |
| $x$ divided by five            | $\frac{x}{5}$ |
| the square of $x$ , less three | $x^2 - 3$     |
| twice the sum of $x$ and $y$   | $2(x + y)$    |

"TWICE THE SUM" NEEDS BRACKETS

$2(x + y)$  and  $2x + y$  are different expressions. The bracket is what makes the doubling apply to the whole sum.

### The parts of an expression

An expression is built from **terms** separated by + and −. Each term is a product of a number and some letters.

| EXPRESSION        | TERMS             | COEFFICIENTS      |
|-------------------|-------------------|-------------------|
| $3x + 5y - 7$     | $3x, 5y, -7$      | $3, 5$            |
| $-4a^2 + a$       | $-4a^2, a$        | $-4, 1$           |
| $x - \frac{y}{2}$ | $x, -\frac{y}{2}$ | $1, -\frac{1}{2}$ |

A term with no letter is a **constant**. A coefficient of 1 or −1 is not written:  $a$  means  $1a$  and  $-a$  means  $-1a$ .

THE SIGN BELONGS TO THE TERM

In  $3x - 7y$  the second term is  $-7y$ , with coefficient  $-7$ . Reading the coefficient as 7 loses the sign in every later calculation.

### Evaluating

To **evaluate** an expression, substitute the given numbers and compute, respecting the order of operations.

| EXPRESSION | $X = 3, Y = -2$ | VALUE |
|------------|-----------------|-------|
| $2x + y$   | $6 + (-2)$      | $4$   |
| $x^2 - y$  | $9 - (-2)$      | $11$  |
| $xy$       | $3 \times (-2)$ | $-6$  |
| $2(x + y)$ | $2(1)$          | $2$   |

PUT NEGATIVE VALUES IN BRACKETS

Writing  $x^2 - y$  as  $9 - -2$  invites an error;  $9 - (-2)$  does not. The brackets cost nothing and prevent the commonest slip in substitution.

**02** Worked examples**EXAMPLE 1** Write as expressions: seven more than twice  $x$ ; the square of the sum of  $a$  and  $b$ .

- ① Twice  $x$  is  $2x$ .
- ② Seven more:  $2x + 7$ .
- ③ The sum is  $a + b$ .
- ④ Its square:  $(a + b)^2$ .  
Brackets are essential.

**ANSWER** $2x + 7$  and  $(a + b)^2$ **EXAMPLE 2** Name the terms and coefficients of  $3x + 5y - 7$ .

- ① Terms are separated by  $+$  and  $-$ .
- ②  $3x$ ,  $5y$ ,  $-7$ .
- ③ Coefficients: 3 and 5.
- ④  $-7$  is a constant term.

**ANSWER**

Three terms; coefficients 3 and 5

**EXAMPLE 3** Evaluate  $x^2 - y$  and  $2(x + y)$  at  $x = 3, y = -2$ .

$$1 \quad 3^2 - (-2) = 9 + 2$$

Brackets round the negative.

$$2 \quad = 11$$

$$3 \quad 2(3 + (-2)) = 2(1)$$

$$4 \quad = 2$$

ANSWER

11 and 2

**03** Interactive model

Read six phrases aloud and have the class write the expressions before any of them is discussed; the errors that appear are exactly the ones to teach.

**Substituting values**

$$3x^2 - 5x + 2$$

x 2

value 4

At  $x = 1$  the value is zero; everywhere else the letter is just a number waiting to be put in.

**04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | O'ZBEKCHA        | РУССКИЙ                  |
|----------------------|------------------|--------------------------|
| Algebraic expression | Algebraik ifoda  | Алгебраическое выражение |
| Variable             | O'zgaruvchi      | Переменная               |
| Term                 | Had              | Член                     |
| Coefficient          | Koeffitsiyent    | Коэффициент              |
| Constant             | O'zgarmas        | Постоянная               |
| To evaluate          | Qiymatini topmoq | Вычислить значение       |
| To substitute        | O'rniga qo'yimoq | Подставить               |
| Equation             | Tenglama         | Уравнение                |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.



**05 Quick check****Check yourself**

“Five less than  $x$ ” is:

“Twice the sum of  $x$  and  $y$ ” is:

How many terms in  $3x + 5y - 7$ ?

The coefficient of  $y$  in  $3x - 7y$ :

7

-7

3

1

$x^2 - y$  at  $x = 3, y = -2$ :

7

11

-11

5

A term with no letter is:

a coefficient

a constant

a variable

an equation

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Five more than  $x$

ANSWER  $x + 5$

2. Five less than  $x$

ANSWER  $x - 5$

3. Five times  $x$

ANSWER  $5x$

4.  $x$  divided by five

ANSWER  $\frac{x}{5}$

5. Terms in  $3x + 5y - 7$

ANSWER 3

6. Coefficient of  $x$  in  $3x$

ANSWER 3

7. Coefficient of  $a$  in  $-a$

ANSWER  $-1$

Medium · the standard question

7 PROBLEMS

1. Twice the sum of  $x$  and  $y$

ANSWER  $2(x + y)$

2. Seven more than twice  $x$

ANSWER  $2x + 7$

3. The square of the sum of  $a$  and  $b$

ANSWER  $(a + b)^2$

4. Coefficient of  $y$  in  $3x - 7y$

ANSWER  $-7$

5.  $2x + y$  at  $x = 3, y = -2$

ANSWER  $4$

6.  $x^2 - y$  at the same

ANSWER  $11$

7.  $xy$  at the same

ANSWER  $-6$

### Hard · stretch and reason

7 PROBLEMS

1.  $x^2 + 2xy + y^2$  at  $x = 3, y = -2$

ANSWER 1

2.  $\frac{x+y}{x-y}$  at  $x = 3, y = -2$

ANSWER  $\frac{1}{5}$

3.  $-a^2$  at  $a = -4$

ANSWER -16

4.  $(-a)^2$  at  $a = -4$

ANSWER 16

5. Write: the sum of the squares of  $a$  and  $b$

ANSWER  $a^2 + b^2$

6. Write: the square of the sum of  $a$  and  $b$

ANSWER  $(a + b)^2$

7. Are they the same?

ANSWER No — try  $a = b = 1$

## 07 Homework

### Homework — 5 tasks

Put every substituted negative value in brackets.

- Write as expressions: four less than three times  $x$ ; half the sum of  $a$  and  $b$ .
- Name the terms and coefficients of  $5a - 2b + 9$ .
- Evaluate  $3x - 2y$  at  $x = -1, y = 4$ .
- Evaluate  $(x + y)^2$  at the same values.

5. Explain the difference between  $a^2 + b^2$  and  $(a + b)^2$ .

# Rules for removing brackets, and the coefficient

A plus leaves the signs alone; a minus changes every one of them.

LESSON 59

1 × 40 MIN

MATEMATIKA 7, §25

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

|                                 |        |
|---------------------------------|--------|
| Plus and minus before a bracket | 10 min |
| A number before a bracket       | 12 min |
| The coefficient                 | 12 min |
| Homework                        | 6 min  |

LEARNING OBJECTIVES

- Remove brackets preceded by a plus and by a minus sign.
- Multiply a bracket by a number using the distributive law.
- Identify and use the coefficient of a term.
- Simplify an expression by removing brackets and collecting terms.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 156–159        |
| Cambridge | Stage 8, pp. 16–20 |
| Workbook  | Exercise 2.2       |

01

## Explanation

### Plus and minus before a bracket

$$a + (b - c) = a + b - c$$
$$a - (b - c) = a - b + c$$

| EXPRESSION      | BRACKETS REMOVED |
|-----------------|------------------|
| $5 + (x - 3)$   | $5 + x - 3$      |
| $5 - (x - 3)$   | $5 - x + 3$      |
| $-(2a - b + 4)$ | $-2a + b - 4$    |
| $7 - (-x + 2)$  | $7 + x - 2$      |

A MINUS CHANGES EVERY SIGN INSIDE

$-(2a - b + 4) = -2a + b - 4$  — all three. Changing only the first is the error this lesson exists to prevent, and it recurs all year.

### A number before a bracket

$$k(a + b) = ka + kb$$

Every term inside is multiplied — including the last one, and including the signs.

| EXPRESSION       | EXPANDED        |
|------------------|-----------------|
| $3(x + 4)$       | $3x + 12$       |
| $3(x - 4)$       | $3x - 12$       |
| $-3(x - 4)$      | $-3x + 12$      |
| $-2(3a - b + 5)$ | $-6a + 2b - 10$ |

$- (...)$  IS  $-1(...)$

The two rules of this lesson are one rule: a minus in front is multiplication by  $-1$ , and the distributive law does the rest.

### The coefficient

The **coefficient** is the numerical factor of a term, taken with its sign.

| TERM           | COEFFICIENT   | LETTERS |
|----------------|---------------|---------|
| $7ab$          | $7$           | $ab$    |
| $-3x^2y$       | $-3$          | $x^2y$  |
| $a$            | $1$           | $a$     |
| $-b$           | $-1$          | $b$     |
| $\frac{2x}{3}$ | $\frac{2}{3}$ | $x$     |

Terms with the **same letters to the same powers** are **like terms**, and only like terms can be collected:  $3x + 5x = 8x$ , but  $3x + 5y$  cannot be simplified.

**$3x$  AND  $3x^2$  ARE NOT LIKE TERMS**

The powers must match too. Adding them gives  $3x + 3x^2$ , which is already as simple as it gets.



## 02 Worked examples

**EXAMPLE 1** Remove the brackets:  $5 - (x - 3)$  and  $-(2a - b + 4)$ .

① A minus before the bracket changes every sign.

②  $5 - x + 3 = 8 - x$

③  $-2a + b - 4$   
All three signs.

④ Nothing further to collect in the second.

ANSWER

$8 - x$  and  $-2a + b - 4$

**EXAMPLE 2** Expand  $-2(3a - b + 5)$ .

① Multiply every term by  $-2$ .

②  $-2 \times 3a = -6a$

③  $-2 \times (-b) = +2b$

④  $-2 \times 5 = -10$ , giving  $-6a + 2b - 10$ .

ANSWER

$-6a + 2b - 10$

**EXAMPLE 3** Simplify  $4(x + 2) - 3(x - 1)$ .

$$① \quad 4x + 8$$

$$② \quad -3x + 3$$

Both signs change.

$$③ \quad 4x + 8 - 3x + 3$$

$$④ \quad = x + 11$$

ANSWER

$$x + 11$$

**03** Interactive model

Write  $-(a - b)$  on the board and ask for its value at  $a = 5$ ,  $b = 2$  before removing the brackets; the numerical check settles the sign rule.

**Removing brackets**

$$5 - (2x - 3) = 8 - 2x$$

x 2

value 4

The minus in front changes both signs inside. Substitute any value into either form and the two agree.

**04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA         | РУССКИЙ                 |
|--------------------|-------------------|-------------------------|
| Bracket            | Qavs              | Скобка                  |
| To remove brackets | Qavslarni ochish  | Раскрыть скобки         |
| Distributive law   | Taqsimot qonuni   | Распределительный закон |
| Coefficient        | Koeffitsiyent     | Коэффициент             |
| Like terms         | O'xshash hadlar   | Подобные члены          |
| To simplify        | Soddalashtirish   | Упростить               |
| Sign change        | Ishora o'zgarishi | Изменение знака         |
| Nested brackets    | Ichma-ich qavslar | Вложенные скобки        |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$5 - (x - 3)$  equals:

$-(2a - b + 4)$  equals:

$-3(x - 4)$  equals:

The coefficient of  $-3x^2y$ :

The coefficient of  $-b$ :

Are  $3x$  and  $3x^2$  like terms?

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $5 + (x - 3)$

ANSWER  $x + 2$ 

2.  $5 - (x - 3)$

ANSWER  $8 - x$ 

3.  $3(x + 4)$

ANSWER  $3x + 12$ 

4.  $3(x - 4)$

ANSWER  $3x - 12$ 

5.  $-3(x - 4)$

ANSWER  $-3x + 12$ 

6. Coefficient of  $7ab$

ANSWER  $7$ 

7. Coefficient of  $-b$

ANSWER  $-1$ **Medium · the standard question**

7 PROBLEMS

1.  $-(2a - b + 4)$

ANSWER  $-2a + b - 4$ 

2.  $-2(3a - b + 5)$

ANSWER  $-6a + 2b - 10$ 

3.  $7 - (-x + 2)$

ANSWER  $x + 5$ 

4.  $4(x + 2) - 3(x - 1)$

ANSWER  $x + 11$ 

5.  $2(a - 3) + 5(a + 1)$

ANSWER  $7a - 1$ 

6. Collect  $3x + 5x - 2x$

ANSWER  $6x$ 

7. Collect  $3x + 5y$

ANSWER Already simplest

**Hard · stretch and reason**

7 PROBLEMS

1.  $3(2x - 1) - 2(3x - 4)$

ANSWER 5

2.  $-[2 - (3 - x)]$

ANSWER  $1 - x$ 

3.  $5 - 2(x - 3(x + 1))$

ANSWER  $4x + 11$ 

4.  $a - (b - (c - d))$

ANSWER  $a - b + c - d$ 

5. Simplify  $4(x + y) - 4(x - y)$

ANSWER  $8y$ 

6. Evaluate  $-(a - b)$  at  $a = 5, b = 2$

ANSWER  $-3$ 

7. Coefficient of  $\frac{3x}{4}$

ANSWER  $\frac{3}{4}$ **07 Homework****Homework — 5 tasks**

Check every expansion by substituting one easy pair of values.

1. Remove the brackets:  $8 - (y - 5)$  and  $-(3m - 2n + 1)$ .
2. Expand  $-4(2a - 3b + 1)$ .
3. Simplify  $5(x - 2) - 2(x + 3)$ .
4. Give the coefficients in  $-5a + b - \frac{c}{2}$ .



5. Simplify  $3(a + b) - 3(a - b)$ .

# Properties of the arithmetic operations

The five laws that make algebra possible, written for letters instead of numbers.

LESSON 60

1 × 40 MIN

MATEMATIKA 7, §26

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

|                   |        |
|-------------------|--------|
| The five laws     | 10 min |
| What is not a law | 12 min |
| Using them        | 12 min |
| Homework          | 6 min  |

LEARNING OBJECTIVES

- State the commutative, associative and distributive laws.
- Use them to reorder and regroup a calculation.
- Recognise that subtraction and division are not commutative.
- Use the laws to simplify an expression quickly.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 160–163        |
| Cambridge | Stage 8, pp. 12–18 |
| Workbook  | Exercise 2.1       |

01

## Explanation

### The five laws

| LAW             | STATEMENT                   | SAYS                       |
|-----------------|-----------------------------|----------------------------|
| commutative (+) | $a + b = b + a$             | order does not matter      |
| commutative (×) | $ab = ba$                   | order does not matter      |
| associative (+) | $(a + b) + c = a + (b + c)$ | grouping does not matter   |
| associative (×) | $(ab)c = a(bc)$             | grouping does not matter   |
| distributive    | $a(b + c) = ab + ac$        | the two operations connect |

Two more, about the special numbers 0 and 1:

$a + 0 = a$  $a \times 1 = a$  $a \times 0 = 0$

THE DISTRIBUTIVE LAW IS THE ONLY ONE THAT MIXES + AND ×

Everything in algebra that turns a sum into a product, or a product into a sum, is that one law — expanding brackets, factorising, and collecting like terms alike.

What is not a law

| CLAIM                               | TRUE? | COUNTER-EXAMPLE                  |
|-------------------------------------|-------|----------------------------------|
| $a - b = b - a$                     | no    | $5 - 3 \neq 3 - 5$               |
| $a \div b = b \div a$               | no    | $6 \div 2 \neq 2 \div 6$         |
| $(a - b) - c = a - (b - c)$         | no    | $(9 - 4) - 2 \neq 9 - (4 - 2)$   |
| $a + (b \times c) = (a + b)(a + c)$ | no    | $2 + 3 \times 4 \neq 5 \times 6$ |

ONE COUNTER-EXAMPLE IS A COMPLETE DISPROOF

A rule that fails once is not a rule. Testing a suspicious claim with small numbers is the fastest check in all of algebra.

Using them

The laws are used to make a calculation easier, not to make it longer.

| CALCULATION             | LAW USED                    | EASY ROUTE                       |
|-------------------------|-----------------------------|----------------------------------|
| $25 \times 17 \times 4$ | commutative and associative | $(25 \times 4) \times 17 = 1700$ |
| $7 \times 98$           | distributive                | $7(100 - 2) = 686$               |
| $36 + 47 + 64$          | commutative                 | $(36 + 64) + 47 = 147$           |
| $4a + 7b - a + 2b$      | commutative                 | $3a + 9b$                        |

LOOK FOR PAIRS THAT MAKE ROUND NUMBERS

25 with 4, 36 with 64, 98 with 100. Spotting them before starting turns a hard calculation into an easy one.

**02** Worked examples**EXAMPLE 1** Compute  $25 \times 17 \times 4$  in the easiest way.

① Reorder:  $25 \times 4 \times 17$ .

Commutative.

② Regroup:  $(25 \times 4) \times 17$ .

Associative.

③  $= 100 \times 17$

④  $= 1700$

**ANSWER** $1700$ **EXAMPLE 2** Compute  $7 \times 98$  without a calculator.

①  $98 = 100 - 2$

②  $7(100 - 2)$

Distributive.

③  $= 700 - 14$

④  $= 686$

**ANSWER** $686$

**EXAMPLE 3** Show that subtraction is not associative.

- ① Test  $(9 - 4) - 2$  and  $9 - (4 - 2)$ .
- ②  $5 - 2 = 3$
- ③  $9 - 2 = 7$
- ④  $3 \neq 7$  — one counter-example is enough.

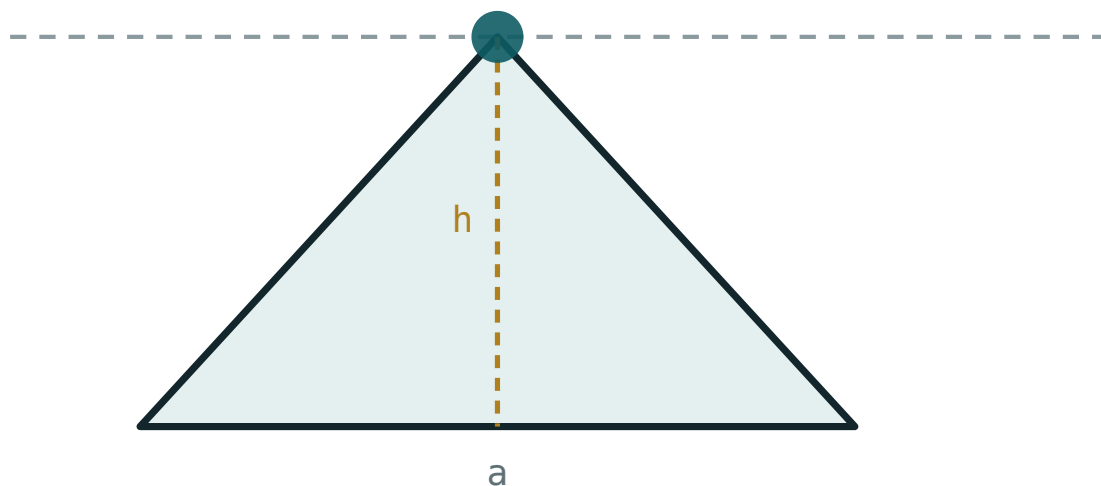
ANSWER

Disproved

**03** Interactive model

Ask for  $25 \times 17 \times 4$  with no calculator; the class that reorders finishes in five seconds and the rest do not.

## The distributive law as an area

base  $a$  **220**height  $h$  **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH          | O'ZBEKCHA                 | РУССКИЙ                 |
|------------------|---------------------------|-------------------------|
| Commutative law  | O'rin almashtirish qonuni | Переместительный закон  |
| Associative law  | Guruhlash qonuni          | Сочетательный закон     |
| Distributive law | Taqsimot qonuni           | Распределительный закон |
| Identity element | Neytral element           | Нейтральный элемент     |
| To reorder       | Tartibini o'zgartirish    | Переставить             |
| To regroup       | Guruhlarni o'zgartirish   | Перегруппировать        |
| Not commutative  | Kommutativ emas           | Не коммутативно         |
| Counter-example  | Qarama-qarshi misol       | Контрпример             |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$a + b = b + a$  is:

associative

commutative

distributive

not a law

$(ab)c = a(bc)$  is:

associative

commutative

distributive

not a law

$a(b + c) = ab + ac$  is:

associative

commutative

distributive

not a law



Is subtraction commutative?

yes

no

sometimes

only for positives

The easiest route to  $25 \times 17 \times 4$ :

left to right

pair 25 with 4

pair 17 with 4

a calculator

One counter-example:

weakens a rule

disproves it

proves it

means nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

**Easy · warm the idea up**

7 PROBLEMS

1. Name the law:  $a + b = b + a$

ANSWER Commutative

2. Name the law:  $(ab)c = a(bc)$

ANSWER Associative

3. Name the law:  $a(b + c) = ab + ac$

ANSWER Distributive

4. Is  $a - b = b - a$ ?

ANSWER No

5. Is  $a \div b = b \div a$ ?

ANSWER No

6.  $a + 0$

ANSWER  $a$

7.  $a \times 1$

ANSWER  $a$

Medium · the standard question

7 PROBLEMS

1.  $25 \times 17 \times 4$

ANSWER 1700

2.  $7 \times 98$

ANSWER 686

3.  $36 + 47 + 64$

ANSWER 147

4.  $8 \times 125 \times 6$

ANSWER 6000

5.  $12 \times 101$

ANSWER 1212

6. Simplify  $4a + 7b - a + 2b$

ANSWER  $3a + 9b$

7. Simplify  $5x - 2y + 3y - x$

ANSWER  $4x + y$

### Hard · stretch and reason

7 PROBLEMS

1.  $99 \times 99 + 99$

ANSWER 9900

2.  $37 \times 25 \times 4$

ANSWER 3700

3.  $16 \times 47 + 16 \times 53$

ANSWER 1600

4. Show that  $(a - b) - c \neq a - (b - c)$

ANSWER  $(9-4)-2 = 3$ ,  $9-(4-2) = 7$

5. Simplify  $2(3a - b) - 3(a - 2b)$

ANSWER  $3a + 4b$

6.  $101 \times 99$

ANSWER 9999

7. Why is the distributive law special?

ANSWER It is the only one linking + and  $\times$

## 07 Homework

### Homework — 5 tasks

Look for the easy pairing before computing anything.

1. Compute  $4 \times 37 \times 25$  using the laws.
2. Compute  $9 \times 102$  using the distributive law.
3. Compute  $58 + 79 + 42$  by reordering.
4. Show with an example that division is not commutative.
5. Simplify  $6a - 2b + 3b - 4a$ .

# Powers with a natural exponent

Repeated multiplication, written short — and the vocabulary that goes with it.

LESSON 61–62

2 × 40 MIN

MATEMATIKA 7, §27

S8 6.1

LESSON CLOCK

13'

23'

22'

18'

4'

|                     |        |
|---------------------|--------|
| The definition      | 13 min |
| Signs and fractions | 23 min |
| Order of operations | 22 min |
| Squares and cubes   | 18 min |
| Homework            | 4 min  |

LEARNING OBJECTIVES

- Define  $a^n$  for a natural exponent and name the base and the exponent.
- Compute powers of positive, negative and fractional bases.
- Use the correct order of operations with powers.
- Recognise squares and cubes of the small numbers.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 164–170        |
| Cambridge | Stage 8, pp. 58–62 |
| Workbook  | Exercise 6.1       |

01

## Explanation

### The definition

$$a^n = a \times a \times \dots \times a \text{ (} n \text{ factors), } n \in \mathbb{N}$$

$a$  is the **base**,  $n$  the **exponent**.  $a^1 = a$ , and  $a^2$  is read “ $a$  squared”,  $a^3$  “ $a$  cubed”.

| POWER  | MEANS                                   | VALUE |
|--------|-----------------------------------------|-------|
| $2^5$  | $2 \times 2 \times 2 \times 2 \times 2$ | 32    |
| $3^4$  | $3 \times 3 \times 3 \times 3$          | 81    |
| $10^3$ | $10 \times 10 \times 10$                | 1000  |
| $7^1$  | 7                                       | 7     |

$2^5$  IS NOT  $2 \times 5$

It is 32, not 10. The exponent counts the factors; it is not a multiplier. Reading the power aloud as “two multiplied by itself five times over” prevents the confusion.

Signs and fractions

| POWER             | VALUE          | WHY                            |
|-------------------|----------------|--------------------------------|
| $(-2)^3$          | $-8$           | three negative factors         |
| $(-2)^4$          | $16$           | four negative factors          |
| $-2^4$            | $-16$          | only the 2 is raised           |
| $(\frac{2}{3})^3$ | $\frac{8}{27}$ | numerator and denominator both |
| $(0.1)^2$         | $0.01$         | $0.1 \times 0.1$               |

$(-a)^n = a^n$  for even  $n$ ,  $(-a)^n = -a^n$  for odd  $n$

THE BRACKETS DECIDE WHAT IS RAISED  
 $(-2)^4 = 16$  but  $-2^4 = -16$ . This distinction is tested every year from now to Grade 11.

Order of operations

Powers come after brackets and before multiplication.

| EXPRESSION           | WORKING      | VALUE |
|----------------------|--------------|-------|
| $3 \times 2^3$       | $3 \times 8$ | $24$  |
| $(3 \times 2)^3$     | $6^3$        | $216$ |
| $2 + 3^2$            | $2 + 9$      | $11$  |
| $(2 + 3)^2$          | $5^2$        | $25$  |
| $5^2 - 3 \times 2^2$ | $25 - 12$    | $13$  |

A BRACKET IS THE ONLY WAY TO RAISE A SUM  
 $(2 + 3)^2$  is 25;  $2 + 3^2$  is 11; and  $2^2 + 3^2$  is 13. Three different expressions, and the brackets are what distinguish them.

Squares and cubes

|       |   |   |    |    |     |     |     |     |     |      |
|-------|---|---|----|----|-----|-----|-----|-----|-----|------|
| $N$   | 1 | 2 | 3  | 4  | 5   | 6   | 7   | 8   | 9   | 10   |
| $n^2$ | 1 | 4 | 9  | 16 | 25  | 36  | 49  | 64  | 81  | 100  |
| $n^3$ | 1 | 8 | 27 | 64 | 125 | 216 | 343 | 512 | 729 | 1000 |

Worth adding:  $2^5 = 32$ ,  $2^6 = 64$ ,  $2^7 = 128$ ,  $2^8 = 256$ ,  $2^{10} = 1024$ , and  $3^4 = 81$ ,  $5^4 = 625$ .

THESE TWO ROWS SAVE MORE TIME THAN ANY RULE

They are used in factorising, in simplifying surds, in Pythagoras and in every prime factorisation. Learning them once repays the effort for five years.

02

Worked examples

EXAMPLE 1 Compute  $(-2)^4$ ,  $-2^4$  and  $(-2)^5$ .

1

$(-2)^4$ : four negative factors — positive.

2

$= 16$

3

$-2^4$  means  $-(2^4)$  — only the 2 is raised.

4

$= -16$ ; and  $(-2)^5 = -32$ .  
Five factors — odd.

ANSWER

16, -16, -32

**EXAMPLE 2** Compute  $5^2 - 3 \times 2^2$ .

- ① Powers first: 25 and 4.
- ② Then multiplication:  $3 \times 4 = 12$ .
- ③ Then subtraction.
- ④  $25 - 12 = 13$

ANSWER

13

**EXAMPLE 3** Compute  $(\frac{2}{3})^3$  and  $(0.1)^2$ .

- ① Raise the numerator and the denominator.
- ②  $\frac{2^3}{3^3} = \frac{8}{27}$
- ③  $0.1 \times 0.1$
- ④  $= 0.01$

ANSWER

 $\frac{8}{27}$  and 0.01**03 Interactive model**

Fold a sheet of paper in half repeatedly and count the layers; the class reaches  $2^7$  before the paper stops folding, and the growth is felt rather than stated.



Powers of a number

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value<sup>3</sup> **64**

a<sup>1</sup> **64**

Raising the answer to the power 3 gives back a<sup>1</sup> — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                 | ЎЗБЕКЧА               | РУССКИЙ             |
|-------------------------|-----------------------|---------------------|
| Power                   | Daraja                | Степень             |
| Base                    | Asos                  | Основание           |
| Exponent                | Ko'rsatkich           | Показатель          |
| Square                  | Kvadrat               | Квадрат             |
| Cube                    | Kub                   | Куб                 |
| To raise to a power     | Darajaga ko'tarish    | Возвести в степень  |
| Repeated multiplication | Takroriy ko'paytirish | Повторное умножение |
| Order of operations     | Amallar tartibi       | Порядок действий    |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$2^5$  equals:

In  $a^n$ ,  $n$  is the:

$(-2)^4$  equals:

$-2^4$  equals:

16

-16

8

-8

$(2 + 3)^2$  equals:

11

13

25

36

$(\frac{2}{3})^3$  equals:

$\frac{2}{27}$

$\frac{8}{27}$

$\frac{6}{9}$

$\frac{8}{3}$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

## Easy · warm the idea up

7 PROBLEMS

1.  $2^5$

ANSWER 32

2.  $3^4$

ANSWER 81

3.  $10^3$

ANSWER 1000

4.  $7^2$

ANSWER 49

5.  $5^3$

ANSWER 125

6.  $(-2)^3$

ANSWER -8

7.  $(-2)^4$

ANSWER 16

## Medium · the standard question

7 PROBLEMS

1.  $-2^4$

ANSWER  $-16$ 

2.  $(\frac{2}{3})^3$

ANSWER  $\frac{8}{27}$ 

3.  $(0.1)^2$

ANSWER  $0.01$ 

4.  $3 \times 2^3$

ANSWER  $24$ 

5.  $(3 \times 2)^3$

ANSWER  $216$ 

6.  $2 + 3^2$

ANSWER  $11$ 

7.  $5^2 - 3 \times 2^2$

ANSWER  $13$ **Hard · stretch and reason**

7 PROBLEMS

1.  $(2 + 3)^2 - 2^2 - 3^2$

ANSWER 12

2.  $(-3)^2 + (-3)^3$

ANSWER -18

3.  $2^8 \div 2^5$

ANSWER 8

4.  $(0.2)^3$

ANSWER 0.008

5.  $(-\frac{1}{2})^4$

ANSWER  $\frac{1}{16}$ 

6. Which is larger,  $2^{10}$  or  $10^3$ ?

ANSWER  $2^{10} = 1024$ 

7.  $3^2 \times 3^3$

ANSWER 243

**07 Homework****Homework — 5 tasks**

Write out the factors for any power you are unsure of.

1. Compute  $2^6$ ,  $3^5$  and  $10^4$ .

2. Compute  $(-3)^3$ ,  $(-3)^4$  and  $-3^4$ .

3. Compute  $(\frac{3}{4})^2$  and  $(0.3)^2$ .

4. Compute  $4^2 + 2 \times 3^2$ .

5. Learn the squares to 12 and the cubes to 6, and write them out from memory.

# Properties of powers with a natural exponent

Five laws — and every one of them is just counting the factors.

LESSON 63–64

2 × 40 MIN

MATEMATIKA 7, §28

S8 6.2

LESSON CLOCK

13'

23'

22'

18'

4'

|                          |        |
|--------------------------|--------|
| Multiplying and dividing | 13 min |
| Powers of powers         | 23 min |
| Products and quotients   | 22 min |
| Putting them together    | 18 min |
| Homework                 | 4 min  |

LEARNING OBJECTIVES

- State and prove the laws of indices for natural exponents.
- Multiply and divide powers with the same base.
- Raise a power, a product and a quotient to a power.
- Simplify an expression containing several powers.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 171–177        |
| Cambridge | Stage 8, pp. 60–65 |
| Workbook  | Exercise 6.2       |

01

## Explanation

### Multiplying and dividing

$$a^m \cdot a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (m > n)$$

**Why.**  $a^3 \cdot a^2 = (aaa)(aa) = a^5$ : the exponents count the factors, and multiplying puts the two lists end to end.

| EXPRESSION      | WORKING   | ANSWER      |
|-----------------|-----------|-------------|
| $2^3 \cdot 2^4$ | $2^{3+4}$ | $2^7 = 128$ |
| $x^5 \cdot x$   | $x^{5+1}$ | $x^6$       |
| $3^7 \div 3^4$  | $3^{7-4}$ | $3^3 = 27$  |
| $a^6 \div a^6$  | $a^0$     | $1$         |

THE BASES MUST BE THE SAME

$2^3 \cdot 3^2$  cannot be combined — the laws say nothing about it. Only  $2^3 \cdot 2^2$  becomes  $2^5$ .



## Powers of powers

$$(a^m)^n = a^{mn}$$

**Why.**  $(a^3)^2 = a^3 \cdot a^3 = a^6$ :  $n$  copies of  $m$  factors is  $mn$  factors.

| EXPRESSION | ANSWER     |
|------------|------------|
| $(2^3)^2$  | $2^6 = 64$ |
| $(x^4)^3$  | $x^{12}$   |
| $(a^2)^5$  | $a^{10}$   |

$(A^3)^2$  AND  $A^3 \cdot A^2$  ARE DIFFERENT

The first is  $a^6$ , the second  $a^5$ . Multiply the exponents for a power of a power; add them for a product.

## Products and quotients

$$(ab)^n = a^n b^n \left( \frac{a}{b} \right)^n = \frac{a^n}{b^n}$$

| EXPRESSION                     | ANSWER           |
|--------------------------------|------------------|
| $(2x)^3$                       | $8x^3$           |
| $(-3a)^2$                      | $9a^2$           |
| $\left( \frac{x}{2} \right)^4$ | $\frac{x^4}{16}$ |
| $(2 \times 5)^3$               | $1000$           |

THERE IS NO SUCH LAW FOR A SUM

$(a + b)^2 \neq a^2 + b^2$ . Try  $a = b = 1$ :  $4 \neq 2$ . The correct expansion is the subject of Quarter III.

## Putting them together

| EXPRESSION               | WORKING              | ANSWER   |
|--------------------------|----------------------|----------|
| $(2x^2)^3$               | $2^3(x^2)^3$         | $8x^6$   |
| $x^5 \cdot x^3 \div x^6$ | $x^{8-6}$            | $x^2$    |
| $(a^3b)^2 \cdot a$       | $a^6b^2 \cdot a$     | $a^7b^2$ |
| $\frac{(3a^2)^3}{9a^4}$  | $\frac{27a^6}{9a^4}$ | $3a^2$   |

DEAL WITH EACH BASE SEPARATELY

Collect the numbers, then the *as*, then the *bs*. Trying to do all three at once is where the errors come from.

02 Worked examples

EXAMPLE 1 Simplify  $2^3 \cdot 2^4$ ,  $3^7 \div 3^4$  and  $(2^3)^2$ .

- 1

Same base, multiply: add the exponents.
- 2

$2^7 = 128$
- 3

Same base, divide: subtract.

$3^3 = 27$ .
- 4

Power of a power: multiply.  $2^6 = 64$ .

ANSWER  
 $128, 27, 64$

**EXAMPLE 2** Simplify  $(2x^2)^3$ .

① Raise each factor:  $2^3 \cdot (x^2)^3$ .

②  $2^3 = 8$

③  $(x^2)^3 = x^6$

Multiply the exponents.

④  $= 8x^6$

ANSWER

$8x^6$

**EXAMPLE 3** Simplify  $\frac{(3a^2)^3}{9a^4}$ .

① Numerator:  $27a^6$ .

② Numbers:  $\frac{27}{9} = 3$ .

③ Letters:  $a^6 \div a^4 = a^2$ .

④  $= 3a^2$

ANSWER

$3a^2$

**03** Interactive model

Write  $a^3 \cdot a^2$  out in full as aaaaa before stating the law; the class sees that the rule is only counting.

Combining powers

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value<sup>3</sup> **64**

a<sup>1</sup> **64**

Raising the answer to the power 3 gives back a<sup>1</sup> — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | ЎЗБЕКЧА                  | РУССКИЙ              |
|--------------------|--------------------------|----------------------|
| Law of indices     | Daraja qonuni            | Свойство степени     |
| Same base          | Bir xil asos             | Одинаковое основание |
| To multiply powers | Darajalarni ko‘paytirish | Умножение степеней   |
| To divide powers   | Darajalarni bo‘lish      | Деление степеней     |
| Power of a power   | Daraja darajasi          | Степень степени      |
| Power of a product | Ko‘paytma darajasi       | Степень произведения |
| Exponent zero      | Nolinch daraja           | Нулевая степень      |
| Simplify           | Soddalashtirish          | Упростить            |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself** $a^m \cdot a^n$  equals:

$a^{mn}$

$a^{m+n}$

$a^{m-n}$

$2a^m$

 $(a^m)^n$  equals:

$a^{m+n}$

$a^{mn}$

$a^{m-n}$

$a^m$

 $3^7 \div 3^4$  equals:

$3^3$

$3^{11}$

$3^{28}$

$1$

$(2x)^3$  equals:

$(a + b)^2$  equals:

$a^6 \div a^6$  equals:

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $2^3 \cdot 2^4$

ANSWER  $2^7$ 

2.  $x^5 \cdot x$

ANSWER  $x^6$ 

3.  $3^7 \div 3^4$

ANSWER  $3^3$ 

4.  $(2^3)^2$

ANSWER  $2^6$ 

5.  $(x^4)^3$

ANSWER  $x^{12}$ 

6.  $(2x)^3$

ANSWER  $8x^3$ 

7.  $a^6 \div a^6$

ANSWER  $1$ **Medium · the standard question**

7 PROBLEMS

1.  $(2x^2)^3$

ANSWER  $8x^6$ 

2.  $(-3a)^2$

ANSWER  $9a^2$ 

3.  $(\frac{x}{2})^4$

ANSWER  $\frac{x^4}{16}$ 

4.  $x^5 \cdot x^3 \div x^6$

ANSWER  $x^2$ 

5.  $(a^3b)^2 \cdot a$

ANSWER  $a^7b^2$ 

6.  $\frac{(3a^2)^3}{9a^4}$

ANSWER  $3a^2$ 

7.  $2^3 \cdot 3^2$

ANSWER 72 — bases differ

**Hard · stretch and reason**

7 PROBLEMS



1. 
$$\frac{(2a^2b)^3}{4a^3b}$$

ANSWER  $2a^3b^2$

2.  $(x^2y^3)^2 \cdot (xy)^3$

ANSWER  $x^7y^9$

3. 
$$\frac{5^{n+1}}{5^n}$$

ANSWER 5

4.  $(-2a^2)^3 \div (4a^3)$

ANSWER  $-2a^3$

5. Is  $(a + b)^2 = a^2 + b^2$ ?

ANSWER No — try  $a = b = 1$

6.  $2^{10} \div 2^7$

ANSWER 8

7.  $(3x)^2 \cdot (2x)^3$

ANSWER  $72x^5$

## 07 Homework

### Homework — 5 tasks

Deal with the numbers first, then each letter separately.

1. Simplify  $5^2 \cdot 5^3$ ,  $7^6 \div 7^4$  and  $(4^2)^3$ .

2. Simplify  $(3x^3)^2$  and  $(-2a)^4$ .

3. Simplify  $y^7 \cdot y^2 \div y^5$ .

4. Simplify  $\frac{(2a^3)^4}{8a^5}$ .

5. Show with numbers that  $(a + b)^2 \neq a^2 + b^2$ .

# The monomial and its standard form

A number times some letters — the simplest object in algebra.

LESSON 65

1 × 40 MIN

MATEMATIKA 7, §29

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

|                       |        |
|-----------------------|--------|
| What a monomial is    | 10 min |
| Standard form         | 12 min |
| Degree and similarity | 12 min |
| Homework              | 6 min  |

LEARNING OBJECTIVES

- Define a monomial and recognise one.
- Write a monomial in standard form.
- Find the coefficient and the degree of a monomial.
- Recognise similar monomials.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 178–181        |
| Cambridge | Stage 8, pp. 12–18 |
| Workbook  | Exercise 2.1       |

01

## Explanation

### What a monomial is

A **monomial** is a product of numbers and letters, with natural-number exponents. No addition, no subtraction and no division by a letter.

| EXPRESSION    | A MONOMIAL? | WHY                  |
|---------------|-------------|----------------------|
| $7a^2b$       | yes         | a product            |
| $-x$          | yes         | $-1 \times x$        |
| $5$           | yes         | a number alone       |
| $a + b$       | no          | a sum                |
| $\frac{3}{x}$ | no          | division by a letter |
| $\frac{x}{3}$ | yes         | $\frac{1}{3}x$       |

DIVIDING BY A NUMBER IS ALLOWED; DIVIDING BY A LETTER IS NOT

$\frac{x}{3}$  is the monomial  $\frac{1}{3}x$ ;  $\frac{3}{x}$  is not a monomial at all. The difference matters for everything that follows.

Standard form

A monomial is in **standard form** when it is written as one number followed by each letter once, in alphabetical order, with its exponent.

| AS WRITTEN                   | STANDARD FORM |
|------------------------------|---------------|
| $3a \cdot 4b$                | $12ab$        |
| $x \cdot 5 \cdot x \cdot y$  | $5x^2y$       |
| $2b \cdot 3a \cdot a$        | $6a^2b$       |
| $-a \cdot a \cdot a \cdot 2$ | $-2a^3$       |

MULTIPLY THE NUMBERS, COLLECT EACH LETTER

Two steps, always in that order. The alphabetical order is only a convention, but it makes two monomials easy to compare.

Degree and similarity

The **degree** of a monomial is the sum of the exponents of its letters. A number alone has degree 0.

| MONOMIAL   | COEFFICIENT | DEGREE |
|------------|-------------|--------|
| $7a^2b$    | 7           | 3      |
| $-5x^3y^2$ | -5          | 5      |
| 4          | 4           | 0      |
| x          | 1           | 1      |

Two monomials are **similar** if their letter parts are identical — same letters, same exponents. Only similar monomials can be added.

| PAIR                 | SIMILAR? |
|----------------------|----------|
| $3a^2b$ and $-7a^2b$ | yes      |
| $3a^2b$ and $3ab^2$  | no       |
| $5x$ and $5x^2$      | no       |

$A^2B$  AND  $AB^2$  ARE NOT SIMILAR

Both have degree 3, but the letters carry different exponents. Similarity is about the letter part, not about the degree.

02

Worked examples

EXAMPLE 1 Write  $2b \cdot 3a \cdot a$  in standard form, and give its coefficient and degree.

1

Numbers:  $2 \times 3 = 6$ .

2

Letters:  $a \cdot a = a^2$ , and one  $b$ .

3

$6a^2b$   
Alphabetical order.

4

Coefficient 6, degree  $2 + 1 = 3$ .

ANSWER

$6a^2b$ ; 6 and 3

**EXAMPLE 2** Which of  $7a^2b$ ,  $a + b$ ,  $\frac{3}{x}$ ,  $\frac{x}{3}$ ,  $5$  are monomials?

- ①  $7a^2b$  — a product ✓
- ②  $a + b$  — a sum ✗
- ③  $\frac{3}{x}$  — division by a letter ✗
- ④  $\frac{x}{3}$  and  $5$  ✓

ANSWER

$7a^2b$ ,  $\frac{x}{3}$ ,  $5$

**EXAMPLE 3** Are  $3a^2b$  and  $3ab^2$  similar? And  $3a^2b$  and  $-7a^2b$ ?

- ① Compare the letter parts.
- ②  $a^2b$  and  $ab^2$  differ.  
Not similar.
- ③  $a^2b$  and  $a^2b$  agree.
- ④ Similar — only the coefficients differ.

ANSWER

No; yes

### 03 Interactive model

Write the same monomial six different ways on the board and ask which is “tidiest”; the class defines standard form before it is named.

Monomials in standard form

$(-3x^2)(2x^3) = -6x^5$

x 2

value -192

Coefficients multiply, exponents add — and the standard form is the one number followed by the letters.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | O‘ZBEKCHA            | РУССКИЙ            |
|----------------------|----------------------|--------------------|
| Monomial             | Bir had              | Одночлен           |
| Standard form        | Standart ko‘rinish   | Стандартный вид    |
| Coefficient          | Koeffitsiyent        | Коэффициент        |
| Degree of a monomial | Bir hadning darajasi | Степень одночлена  |
| Similar monomials    | O‘xshash bir hadlar  | Подобные одночлены |
| Letter part          | Harfiy qism          | Буквенная часть    |
| Alphabetical order   | Alifbo tartibi       | Алфавитный порядок |
| Zero monomial        | Nol bir had          | Нулевой одночлен   |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Which is a monomial?

$$a + b$$

$$7a^2b$$

$$\frac{3}{x}$$

$$a - 1$$

Is  $\frac{x}{3}$  a monomial?

yes

no

sometimes

only if  $x > 0$

Standard form of  $2b \cdot 3a \cdot a$ :

$$6ab^2$$

$$6a^2b$$

$$5a^2b$$

$$6a^3$$



The degree of  $-5x^3y^2$ :

The degree of 4:

Are  $3a^2b$  and  $3ab^2$  similar?

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is  $7a^2b$  a monomial?

ANSWER Yes

2. Is  $a + b$  a monomial?

ANSWER No

3. Is  $\frac{3}{x}$  a monomial?

ANSWER No

4. Is 5 a monomial?

ANSWER Yes

5. Coefficient of  $7a^2b$

ANSWER 7

6. Degree of  $7a^2b$

ANSWER 3

7. Degree of 4

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of  $3a \cdot 4b$

ANSWER  $12ab$

2. Standard form of  $x \cdot 5 \cdot x \cdot y$

ANSWER  $5x^2y$

3. Standard form of  $2b \cdot 3a \cdot a$

ANSWER  $6a^2b$

4. Coefficient and degree of  $-5x^3y^2$

ANSWER  $-5$  and  $5$

5. Are  $3a^2b$  and  $-7a^2b$  similar?

ANSWER Yes

6. Are  $5x$  and  $5x^2$  similar?

ANSWER No

7. Standard form of  $-a \cdot a \cdot a \cdot 2$

ANSWER  $-2a^3$

**Hard · stretch and reason**

7 PROBLEMS

1. Standard form of  $(2a)^3 \cdot 3b$

ANSWER  $24a^3b$

2. Degree of  $(3x^2y)^2$

ANSWER 6

3. Standard form of  $\frac{1}{2}x \cdot 4x^2y$

ANSWER  $2x^3y$

4. Two similar monomials of degree 4

ANSWER  $3a^2b^2$  and  $-a^2b^2$

5. Is  $x^{-1}$  a monomial?

ANSWER No — the exponent is not natural

6. Degree of  $7a^2b \cdot 3ab^3$

ANSWER 7

7. Coefficient of  $-\frac{2x^3}{5}$

ANSWER  $-\frac{2}{5}$

## 07 Homework

### Homework — 5 tasks

Write every monomial in standard form before giving its coefficient or degree.

- Which of  $4xy$ ,  $x + 3$ ,  $\frac{5}{y}$ ,  $-2a^3$ ,  $7$  are monomials?
- Write  $5a \cdot 2b \cdot a$  in standard form.
- Give the coefficient and degree of  $-3x^2y^4$ .

4. Are  $4ab^2$  and  $-ab^2$  similar? And  $4ab^2$  and  $4a^2b$ ?
5. Write  $(3x)^2 \cdot 2y$  in standard form and give its degree.

# Multiplication and division of monomials

Numbers with numbers, letters with letters — and the laws of indices do the rest.

LESSON 66–67

2 × 40 MIN

MATEMATIKA 7, §30

S8 2.1

LESSON CLOCK

13'

23'

22'

18'

4'

|                    |        |
|--------------------|--------|
| Multiplying        | 13 min |
| Raising to a power | 23 min |
| Dividing           | 22 min |
| Areas and volumes  | 18 min |
| Homework           | 4 min  |

LEARNING OBJECTIVES

- Multiply two or more monomials and give the answer in standard form.
- Divide one monomial by another when the division is exact.
- Raise a monomial to a power.
- Use the operations in area and volume problems.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 182–188        |
| Cambridge | Stage 8, pp. 12–18 |
| Workbook  | Exercise 2.1       |

01

## Explanation

### Multiplying

Multiply the coefficients; then, for each letter, add the exponents.

| PRODUCT                 | NUMBERS | LETTERS  | ANSWER      |
|-------------------------|---------|----------|-------------|
| $3a \cdot 4b$           | 12      | $ab$     | $12ab$      |
| $2x^2 \cdot 5x^3$       | 10      | $x^5$    | $10x^5$     |
| $(-3a^2b)(4ab^3)$       | -12     | $a^3b^4$ | $-12a^3b^4$ |
| $(\frac{1}{2}xy)(6x^2)$ | 3       | $x^3y$   | $3x^3y$     |

THE PRODUCT OF TWO MONOMIALS IS ALWAYS A MONOMIAL

Its degree is the sum of the two degrees. That is worth checking as you go:  $(-3a^2b)$  has degree 3,  $(4ab^3)$  degree 4, and the answer degree 7 ✓

Raising to a power

$$(ka^mb^n)^p = k^pa^{mp}b^{np}$$

Every factor is raised — the coefficient included.

| EXPRESSION  | ANSWER    |
|-------------|-----------|
| $(2x)^3$    | $8x^3$    |
| $(3a^2b)^2$ | $9a^4b^2$ |
| $(-2x^3)^3$ | $-8x^9$   |
| $(-2x^3)^2$ | $4x^6$    |

THE COEFFICIENT IS RAISED TOO

$(3a^2b)^2 = 9a^4b^2$ , not  $3a^4b^2$ . Forgetting to square the 3 is the standard slip.

Dividing

Divide the coefficients; then, for each letter, subtract the exponents.

| QUOTIENT                | NUMBERS | LETTERS | ANSWER   |
|-------------------------|---------|---------|----------|
| $12a^5 \div 4a^2$       | 3       | $a^3$   | $3a^3$   |
| $-15x^3y^2 \div 5xy$    | -3      | $x^2y$  | $-3x^2y$ |
| $8a^4b^3 \div (-2a^4b)$ | -4      | $b^2$   | $-4b^2$  |

THE QUOTIENT IS A MONOMIAL ONLY IF EVERY EXPONENT STAYS NATURAL

$a^2 \div a^5$  is not a monomial. In Grade 7 the divisions set are always exact, but the condition is worth knowing.

Areas and volumes

| FIGURE    | DIMENSIONS  | AREA OR VOLUME |
|-----------|-------------|----------------|
| rectangle | $3a$        | $12ab$         |
| square    | $2x$        | $4x^2$         |
| cuboid    | $a, 2a, 3a$ | $6a^3$         |
| cube      | $2x$        | $8x^3$         |

The first row assumes a rectangle  $3a$  by  $4b$ . Notice that the units follow the degree: a length is degree 1, an area degree 2, a volume degree 3.

THE DEGREE IS A DIMENSION CHECK

An answer of degree 2 for a volume is wrong before any arithmetic is examined. This is the algebraic version of checking the units.

02

Worked examples

EXAMPLE 1 Simplify  $(-3a^2b)(4ab^3)$ .

1

Numbers:  $-3 \times 4 = -12$ .

2

$a$ :  $a^2 \cdot a = a^3$ .

3

$b$ :  $b \cdot b^3 = b^4$ .

4

$= -12a^3b^4$   
Degree 7 ✓

ANSWER

$-12a^3b^4$



**EXAMPLE 2** Simplify  $(3a^2b)^2$  and  $(-2x^3)^3$ .

① Square every factor:  $3^2 = 9$ ,  $(a^2)^2 = a^4$ ,  $b^2$ .

②  $= 9a^4b^2$

③ Cube every factor:  $(-2)^3 = -8$ ,  $(x^3)^3 = x^9$ .

④  $= -8x^9$

**ANSWER** $9a^4b^2$  and  $-8x^9$ **EXAMPLE 3** Simplify  $-15x^3y^2 \div 5xy$ .

① Numbers:  $-15 \div 5 = -3$ .

②  $x$ :  $x^3 \div x = x^2$ .

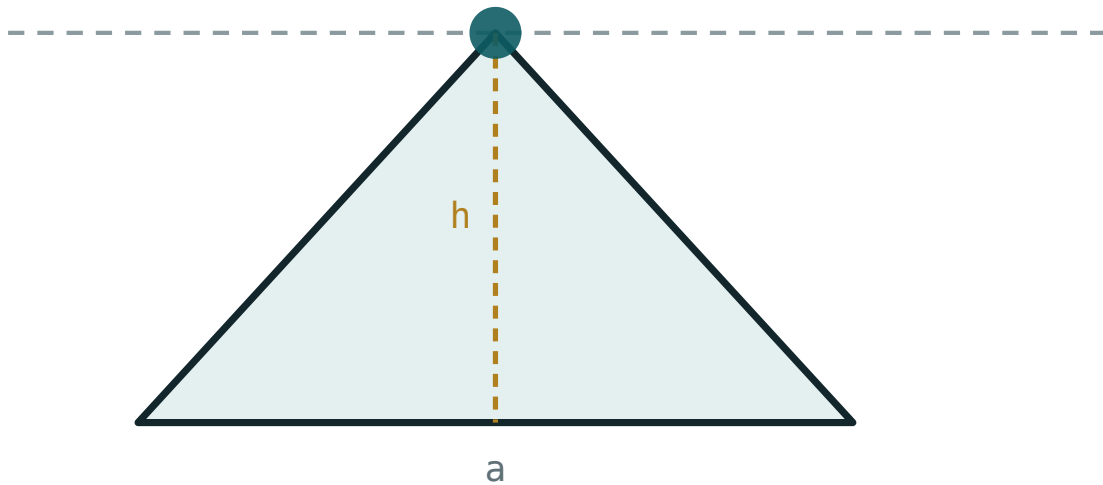
③  $y$ :  $y^2 \div y = y$ .

④  $= -3x^2y$

**ANSWER** $-3x^2y$ **03 Interactive model**

Draw a rectangle  $3a$  by  $4b$  on the board and ask for its area before any rule is stated; the class multiplies numbers and letters separately without being told to.

### A rectangle with algebraic sides

base  $a$  **220**height  $h$  **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH             | O'ZBEKCHA          | РУССКИЙ           |
|---------------------|--------------------|-------------------|
| To multiply         | Ko'paytirish       | Умножить          |
| To divide           | Bo'lish            | Разделить         |
| Standard form       | Standart ko'rinish | Стандартный вид   |
| Exactly divisible   | Butun bo'linadi    | Делится нацело    |
| Power of a monomial | Bir had darajasi   | Степень одночлена |
| Area                | Yuza               | Площадь           |
| Volume              | Hajm               | Объём             |
| Cancel              | Qisqartirish       | Сократить         |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself** $2x^2 \cdot 5x^3$  equals:

$10x^5$

$10x^6$

$7x^5$

$10x$

 $(3a^2b)^2$  equals:

$3a^4b^2$

$9a^4b^2$

$9a^4b$

$6a^4b^2$

 $(-2x^3)^3$  equals:

$-8x^9$

$8x^9$

$-6x^9$

$-8x^6$

$12a^5 \div 4a^2$  equals:

$3a^3$

$3a^7$

$8a^3$

$3a^2$

The degree of a product of monomials is:

the larger degree

the sum of the degrees

the product of them

unchanged

A cube of side  $2x$  has volume:

$2x^3$

$6x^3$

$8x^3$

$8x^2$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $3a \cdot 4b$

ANSWER  $12ab$ 

2.  $2x^2 \cdot 5x^3$

ANSWER  $10x^5$ 

3.  $(2x)^3$

ANSWER  $8x^3$ 

4.  $(3a^2b)^2$

ANSWER  $9a^4b^2$ 

5.  $12a^5 \div 4a^2$

ANSWER  $3a^3$ 

6.  $(-2x^3)^2$

ANSWER  $4x^6$ 

7.  $(-2x^3)^3$

ANSWER  $-8x^9$ **Medium · the standard question**

7 PROBLEMS

1.  $(-3a^2b)(4ab^3)$

ANSWER  $-12a^3b^4$ 

2.  $(\frac{1}{2}xy)(6x^2)$

ANSWER  $3x^3y$ 

3.  $-15x^3y^2 \div 5xy$

ANSWER  $-3x^2y$ 

4.  $8a^4b^3 \div (-2a^4b)$

ANSWER  $-4b^2$ 

5. Area of a rectangle  $3a$  by  $4b$

ANSWER  $12ab$ 

6. Volume of a cuboid  $a$ ,  $2a$ ,  $3a$

ANSWER  $6a^3$ 

7. Degree of  $(-3a^2b)(4ab^3)$

ANSWER  $7$ **Hard · stretch and reason****7 PROBLEMS**

1.  $(2a^2b)^3 \cdot 3ab$

ANSWER  $24a^7b^4$ 

2.  $\frac{(3xy^2)^3}{9x^2y^4}$

ANSWER  $3xy^2$ 

3.  $(-a^2b)^4 \div (a^3b^2)$

ANSWER  $a^5b^2$ 

4. A cube of volume  $27x^6$ : its edge

ANSWER  $3x^2$ 

5. A square of area  $49a^4$ : its side

ANSWER  $7a^2$ 

6.  $(2x)^2 \cdot (3x)^3$

ANSWER  $108x^5$ 

7. Why is  $a^2 \div a^5$  not a monomial?

ANSWER The exponent would be negative

**07 Homework****Homework — 5 tasks**

Check the degree of every answer against the degrees you started with.

1. Simplify  $(-2a^3b)(5ab^2)$ .
2. Simplify  $(4x^2y)^2$  and  $(-3a)^3$ .
3. Simplify  $20a^6b^3 \div 5a^2b$ .
4. Find the volume of a cuboid with edges  $2a$ ,  $3a$  and  $4b$ .



5. A cube has volume  $64x^3$ . Find its edge.

# Polynomials

A sum of monomials — and the object that the rest of algebra is about.

LESSON 68

1 × 40 MIN

MATEMATIKA 7, §31

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

|                          |        |
|--------------------------|--------|
| The definition           | 10 min |
| Standard form and degree | 12 min |
| Evaluating               | 12 min |
| Homework                 | 6 min  |

LEARNING OBJECTIVES

- Define a polynomial and name its terms.
- Write a polynomial in standard form and give its degree.
- Name a binomial and a trinomial.
- Evaluate a polynomial for given values.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 189–192        |
| Cambridge | Stage 8, pp. 16–20 |
| Workbook  | Exercise 2.2       |

01

## Explanation

### The definition

A **polynomial** is a sum of monomials. Each monomial is a **term**.

| POLYNOMIAL          | TERMS | NAME       |
|---------------------|-------|------------|
| $3x$                | 1     | monomial   |
| $3x + 5$            | 2     | binomial   |
| $x^2 - 4x + 3$      | 3     | trinomial  |
| $a^3 + a^2 - a + 7$ | 4     | polynomial |

Subtraction counts as addition of a negative, so  $x^2 - 4x$  is the sum of  $x^2$  and  $-4x$ .

EVERY MONOMIAL IS A POLYNOMIAL

With one term. The names simply record how many terms there are, and only the first three have special names.

### Standard form and degree

A polynomial is in **standard form** when every term is a monomial in standard form, no two terms are similar, and the terms are written in decreasing degree.

| AS WRITTEN        | STANDARD FORM    | DEGREE |
|-------------------|------------------|--------|
| $5 + x^2 - 3x$    | $x^2 - 3x + 5$   | 2      |
| $2x + 3x$         | $5x$             | 1      |
| $a^3 + 2a^3 - 4$  | $3a^3 - 4$       | 3      |
| $x^2y + xy^2 + 1$ | already standard | 3      |

The **degree of a polynomial** is the greatest of the degrees of its terms. The term of that degree is the **leading term**; a term with no letters is the **constant term**.

COLLECT SIMILAR TERMS BEFORE READING THE DEGREE

In  $a^3 + 2a^3 - 4$  the first two terms combine. Reading the degree from an uncollected expression can give the wrong answer when terms cancel.

Evaluating

Substitute and compute, with every negative value in brackets.

| POLYNOMIAL     | $X = 2$          | $X = -1$        |
|----------------|------------------|-----------------|
| $x^2 - 4x + 3$ | $4 - 8 + 3 = -1$ | $1 + 4 + 3 = 8$ |
| $2x^3 - x$     | $16 - 2 = 14$    | $-2 + 1 = -1$   |
| $5 - x^2$      | $5 - 4 = 1$      | $5 - 1 = 4$     |

A VALUE THAT MAKES THE POLYNOMIAL ZERO IS A ROOT

$x^2 - 4x + 3$  is zero at  $x = 1$  and at  $x = 3$ . Finding roots is what the whole of the equation work later this year is about.

**02** Worked examples**EXAMPLE 1** Write  $5 + x^2 - 3x$  in standard form and give its degree and constant term.

- 1 Order by decreasing degree.  
-----
- 2  $x^2 - 3x + 5$   
-----
- 3 Degree 2 — the greatest term degree.  
-----
- 4 Constant term 5.

**ANSWER** $x^2 - 3x + 5$ ; degree 2, constant 5**EXAMPLE 2** Simplify  $a^3 + 2a^3 - 4$  and give its degree.

- 1 Collect the similar terms.  
-----
- 2  $3a^3 - 4$   
-----
- 3 Two terms — a binomial.  
-----
- 4 Degree 3.

**ANSWER** $3a^3 - 4$ , degree 3

**EXAMPLE 3** Evaluate  $x^2 - 4x + 3$  at  $x = 2$  and at  $x = -1$ .

①  $4 - 8 + 3$

②  $= -1$

③  $(-1)^2 - 4(-1) + 3 = 1 + 4 + 3$

Brackets round the negative.

④  $= 8$

ANSWER

$-1$  and  $8$

### 03 Interactive model

Ask for the value of  $x^2 - 4x + 3$  at  $x = 1$  and  $x = 3$ ; both give zero, and the idea of a root arrives before any equation is written.

#### Evaluating a polynomial

$$2x^3 - 5x^2 + 3x - 7$$

x 2

value -5

A polynomial is a machine: put a number in, get a number out. Nothing else about it changes.

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                | O'ZBEKCHA          | РУССКИЙ            |
|------------------------|--------------------|--------------------|
| Polynomial             | Ko'phad            | Многочлен          |
| Binomial               | Ikki had           | Двучлен            |
| Trinomial              | Uch had            | Трёхчлен           |
| Term of a polynomial   | Ko'phad hadi       | Член многочлена    |
| Degree of a polynomial | Ko'phad darajasi   | Степень многочлена |
| Standard form          | Standart ko'rinish | Стандартный вид    |
| Leading term           | Bosh had           | Старший член       |
| Constant term          | Ozod had           | Свободный член     |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A polynomial is:

a product of monomials

a sum of monomials

a fraction

an equation

$3x + 5$  is a:

monomial

binomial

trinomial

constant

The degree of  $x^2 - 3x + 5$ :

1

2

3

5

The constant term of  $x^2 - 3x + 5$ :

$a^3 + 2a^3 - 4$  simplifies to:

$x^2 - 4x + 3$  at  $x = -1$ :

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1. How many terms in  $3x + 5$ ?

ANSWER 2

2. Its name

ANSWER Binomial

3. How many terms in  $x^2 - 4x + 3$ ?

ANSWER 3

4. Its name

ANSWER Trinomial

5. Degree of  $x^2 - 3x + 5$

ANSWER 2

6. Constant term of  $x^2 - 3x + 5$

ANSWER 5

7. Degree of 7

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of  $5 + x^2 - 3x$

ANSWER  $x^2 - 3x + 5$

2. Simplify  $2x + 3x$

ANSWER  $5x$

3. Simplify  $a^3 + 2a^3 - 4$

ANSWER  $3a^3 - 4$

4. Degree of  $x^2y + xy^2 + 1$

ANSWER  $3$

5.  $x^2 - 4x + 3$  at  $x = 2$

ANSWER  $-1$

6.  $x^2 - 4x + 3$  at  $x = -1$

ANSWER  $8$

7.  $2x^3 - x$  at  $x = 2$

ANSWER  $14$

**Hard · stretch and reason**

7 PROBLEMS

1. For which  $x$  is  $x^2 - 4x + 3 = 0$ ?

ANSWER  $x = 1$  and  $x = 3$

2. Simplify  $3x^2 - 5x + 2 - x^2 + 5x - 7$

ANSWER  $2x^2 - 5$

3. Its degree

ANSWER 2

4. Degree of  $(x + 1)(x - 1)$

ANSWER 2

5.  $5 - x^2$  at  $x = -3$

ANSWER -4

6. A polynomial of degree 0

ANSWER A non-zero constant

7. Simplify  $a^2 + a^2 + a^2$

ANSWER  $3a^2$

## 07 Homework

### Homework — 5 tasks

Collect similar terms before naming the degree.

1. Write  $7 - 2x + x^3$  in standard form and give its degree.
2. Simplify  $4y^2 + 3y^2 - 5$  and name the result.
3. Give the constant term of  $2x^2 - x + 9$ .
4. Evaluate  $x^2 + 2x - 5$  at  $x = 3$  and at  $x = -2$ .
5. Find the value of  $x$  for which  $3x - 6 = 0$ .

# Like terms and their collection

Only terms with the same letters can be added — and that one rule tidies every expression.

LESSON 69–70

2 × 40 MIN

MATEMATIKA 7, §32

S8 2.2

LESSON CLOCK

13'

23'

22'

18'

4'

|                       |        |
|-----------------------|--------|
| Which terms are alike | 13 min |
| Collecting            | 23 min |
| After brackets        | 22 min |
| In problems           | 18 min |
| Homework              | 4 min  |

LEARNING OBJECTIVES

- Recognise like terms.
- Collect like terms, including with negative coefficients.
- Simplify an expression after removing brackets.
- Use collection to shorten a perimeter or a cost calculation.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 193–198        |
| Cambridge | Stage 8, pp. 16–20 |
| Workbook  | Exercise 2.2       |

01

## Explanation

### Which terms are alike

Two terms are **like** if their letter parts are identical — the same letters with the same exponents. The coefficients may be anything.

| PAIR           | LIKE? | WHY                  |
|----------------|-------|----------------------|
| $3x, -7x$      | yes   | same letter part $x$ |
| $4a^2b, a^2b$  | yes   | same letter part     |
| $4a^2b, 4ab^2$ | no    | different exponents  |
| $5x, 5y$       | no    | different letters    |
| $7, -2$        | yes   | both constants       |

THINK OF THE LETTER PART AS A UNIT

Three apples plus five apples is eight apples; three apples plus five pears is neither. That is the whole rule, and it explains why unlike terms simply stay as they are.

Collecting

$$ka + la = (k + l)a$$

Add the coefficients and keep the letter part.

| EXPRESSION            | COLLECTED  |
|-----------------------|------------|
| $3x + 5x$             | $8x$       |
| $7a - 10a$            | $-3a$      |
| $4x + 3y - x + 5y$    | $3x + 8y$  |
| $2a^2 + 3a - a^2 + a$ | $a^2 + 4a$ |

TAKE THE SIGN WITH THE TERM

In  $4x + 3y - x + 5y$  the third term is  $-x$ , so the  $x$  terms give  $4 - 1 = 3$ . Copying a term without its sign is the commonest error here.

After brackets

Remove the brackets first, then collect.

| EXPRESSION            | BRACKETS REMOVED    | COLLECTED |
|-----------------------|---------------------|-----------|
| $2(x + 3) + 3(x - 1)$ | $2x + 6 + 3x - 3$   | $5x + 3$  |
| $5(a - 2) - 2(a - 5)$ | $5a - 10 - 2a + 10$ | $3a$      |
| $x - (3x - 4)$        | $x - 3x + 4$        | $-2x + 4$ |

TWO PASSES, NEVER ONE

Removing brackets and collecting at the same time is where the signs go wrong. Write the expanded line in full before touching the collection.

In problems

| SITUATION                                     | EXPRESSION       | SIMPLIFIED |
|-----------------------------------------------|------------------|------------|
| perimeter of a rectangle $a$ by $b$           | $a + b + a + b$  | $2a + 2b$  |
| perimeter of a triangle $x$ , $x$ , $2x - 1$  | $x + x + 2x - 1$ | $4x - 1$   |
| 3 pens at $p$ and 4 books at $b$ , twice over | $2(3p + 4b)$     | $6p + 8b$  |

SIMPLIFYING IS WHAT MAKES A FORMULA USABLE

$2a + 2b$  can be evaluated in two operations;  $a + b + a + b$  needs three. Over a whole calculation the saving is large, and the shorter form is also easier to read.

02 Worked examples

EXAMPLE 1 Collect  $4x + 3y - x + 5y$ .

- 1

$x$  terms:  $4x - x = 3x$ .

The sign belongs to the term.
- 2

$y$  terms:  $3y + 5y = 8y$ .
- 3

They cannot be combined further.
- 4

$3x + 8y$

ANSWER

$3x + 8y$

**EXAMPLE 2** Simplify  $5(a - 2) - 2(a - 5)$ .

- ①  $5a - 10$  and  $-2a + 10$ .  
Both signs change in the second.

②  $5a - 10 - 2a + 10$

③  $a$  terms:  $3a$ ; constants:  $0$ .

④  $= 3a$

ANSWER

$$3a$$

**EXAMPLE 3** Find and simplify the perimeter of a triangle with sides  $x$ ,  $x$  and  $2x - 1$ .

①  $P = x + x + (2x - 1)$

②  $= x + x + 2x - 1$

③ Collect:  $4x - 1$ .

④ Check at  $x = 3$ :  $3 + 3 + 5 = 11 = 4(3) - 1$  ✓

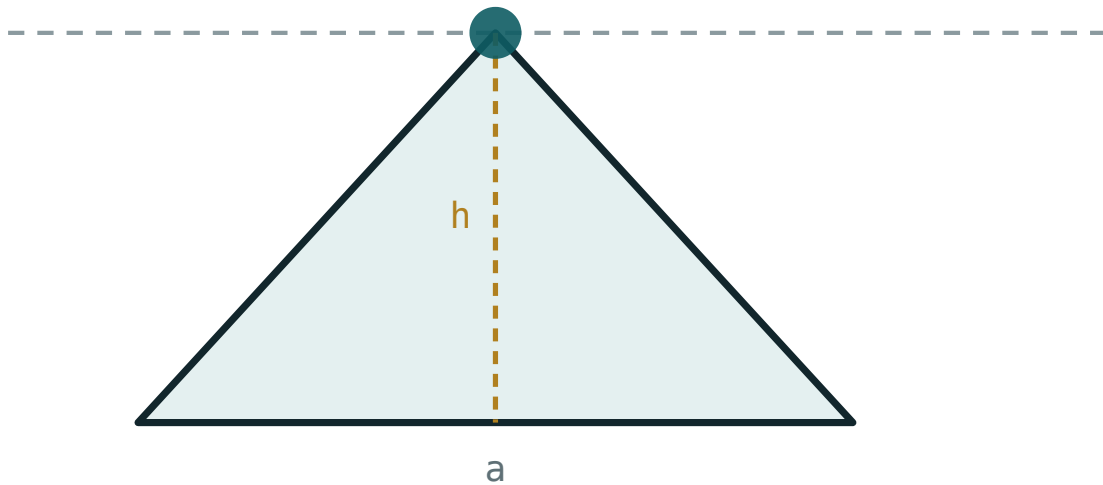
ANSWER

$$4x - 1$$

**03 Interactive model**

Ask for the perimeter of a rectangle twice — once unsimplified and once collected — and have the class evaluate both at  $a = 7$ ,  $b = 4$ ; the answers agree and one is far quicker.

## Collecting like terms

base  $a$  **220**height  $h$  **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**
**04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.



| ENGLISH      | O'ZBEKCHA                 | РУССКИЙ             |
|--------------|---------------------------|---------------------|
| Like terms   | O'xshash hadlar           | Подобные члены      |
| To collect   | Ixchamlash                | Приведение подобных |
| Unlike terms | O'xshash bo'lmagan hadlar | Неподобные члены    |
| Coefficient  | Koeffitsiyent             | Коэффициент         |
| Letter part  | Harfiy qism               | Буквенная часть     |
| To simplify  | Soddalashtirish           | Упростить           |
| Perimeter    | Perimetr                  | Периметр            |
| Cost         | Narx                      | Стоимость           |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Are  $4a^2b$  and  $4ab^2$  like terms?

$3x + 5x$  equals:

$7a - 10a$  equals:

$4x + 3y - x + 5y$  equals:

$3x + 8y$

$5x + 8y$

$3x + 2y$

$11xy$

$5(a - 2) - 2(a - 5)$  equals:

$3a - 20$

$3a$

$7a - 20$

$3a + 20$

Can  $3x$  and  $5y$  be collected?

yes

no

as  $8xy$

as  $15xy$

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $3x + 5x$

ANSWER  $8x$

2.  $7a - 10a$

ANSWER  $-3a$

3.  $2b + b$

ANSWER  $3b$

4.  $5x - 5x$

ANSWER  $0$

5.  $4a^2 + 3a^2$

ANSWER  $7a^2$

6. Can  $3x$  and  $5y$  be collected?

ANSWER No

7. Are  $7$  and  $-2$  like terms?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1.  $4x + 3y - x + 5y$

ANSWER  $3x + 8y$

2.  $2a^2 + 3a - a^2 + a$

ANSWER  $a^2 + 4a$

3.  $2(x + 3) + 3(x - 1)$

ANSWER  $5x + 3$

4.  $5(a - 2) - 2(a - 5)$

ANSWER  $3a$

5.  $x - (3x - 4)$

ANSWER  $4 - 2x$

6. Perimeter of a rectangle  $a$  by  $b$

ANSWER  $2a + 2b$

7. Perimeter of a triangle  $x, x, 2x - 1$

ANSWER  $4x - 1$

**Hard · stretch and reason**

7 PROBLEMS

1.  $3(2a - b) - 2(3a - 2b)$

ANSWER  $b$ 

2.  $4x^2 - 3x + 1 - (x^2 - 3x + 5)$

ANSWER  $3x^2 - 4$ 

3.  $a - [2a - (3a - b)]$

ANSWER  $2a - b$ 

4.  $2(x + y) - 3(x - y) + x$

ANSWER  $5y$ 

5. Perimeter of a square of side  $3x - 2$

ANSWER  $12x - 8$ 

6. A rectangle  $x + 3$  by  $x - 3$ : its perimeter

ANSWER  $4x$ 

7. Simplify  $5ab - 2ba$

ANSWER  $3ab$ 

## 07 Homework

### Homework — 5 tasks

Write the expanded line in full before collecting anything.

1. Collect  $6a + 2b - 4a + 7b$ .

2. Simplify  $3(x + 4) - 2(x - 1)$ .

3. Simplify  $y - (4y - 7)$ .

4. Find and simplify the perimeter of a rectangle  $2x + 1$  by  $x - 2$ .

5. Simplify  $5m^2 + 2m - 3m^2 - 6m$ .

# Addition and subtraction of polynomials

Remove the brackets, collect the like terms — two steps and nothing more.

LESSON 71–72

2 × 40 MIN

MATEMATIKA 7, §33

S8 2.2

LESSON CLOCK

13'

23'

22'

18'

4'

|                    |        |
|--------------------|--------|
| Adding             | 13 min |
| Subtracting        | 23 min |
| Longer expressions | 22 min |
| Checking           | 18 min |
| Homework           | 4 min  |

LEARNING OBJECTIVES

- Add two polynomials.
- Subtract one polynomial from another, changing every sign.
- Simplify an expression containing several polynomials.
- Check an answer by substituting a value.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 199–204        |
| Cambridge | Stage 8, pp. 16–20 |
| Workbook  | Exercise 2.2       |

01

## Explanation

### Adding

A plus before a bracket leaves every sign as it is.

$$(3x^2 - 2x + 5) + (x^2 + 4x - 1) = 4x^2 + 2x + 4$$

| STEP            | LINE                           |
|-----------------|--------------------------------|
| remove brackets | $3x^2 - 2x + 5 + x^2 + 4x - 1$ |
| $x^2$           | $3x^2 + x^2 = 4x^2$            |
| $x$             | $-2x + 4x = 2x$                |
| constants       | $5 - 1 = 4$                    |

WORK ONE DEGREE AT A TIME

All the  $x^2$  terms, then all the  $x$  terms, then the constants. Doing them in a fixed order means nothing is missed.

## Subtracting

A minus before a bracket changes **every** sign inside it.

$$(3x^2 - 2x + 5) - (x^2 + 4x - 1) = 2x^2 - 6x + 6$$

| STEP            | LINE                           |
|-----------------|--------------------------------|
| remove brackets | $3x^2 - 2x + 5 - x^2 - 4x + 1$ |
| $x^2$           | $3x^2 - x^2 = 2x^2$            |
| $x$             | $-2x - 4x = -6x$               |
| constants       | $5 + 1 = 6$                    |

ALL THREE SIGNS CHANGE, NOT JUST THE FIRST

$-(x^2 + 4x - 1)$  is  $-x^2 - 4x + 1$ . Writing that line out in full, before collecting, is the single most useful habit in this topic.

## Longer expressions

| EXPRESSION                     | ANSWER      |
|--------------------------------|-------------|
| $(a + b) + (a - b)$            | $2a$        |
| $(a + b) - (a - b)$            | $2b$        |
| $(2x - 3) + (x + 1) - (x - 4)$ | $2x + 2$    |
| $2(x^2 - x) - 3(x^2 + x)$      | $-x^2 - 5x$ |

The last row combines this lesson with the distributive law: expand first, then collect.

THE DEGREE NEVER RISES

Adding or subtracting polynomials of degree 2 gives a polynomial of degree at most 2 — and sometimes less, if the leading terms cancel.

## Checking

Substitute one convenient value into the original and into the answer. If they differ, the simplification is wrong.



| CHECK                              | ORIGINAL AT $x = 1$ | ANSWER AT $x = 1$           |
|------------------------------------|---------------------|-----------------------------|
| $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$ | $6 + 4 = 10$        | $4 + 2 + 4 = 10 \checkmark$ |
| $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$ | $6 - 4 = 2$         | $2 - 6 + 6 = 2 \checkmark$  |

USE  $x = 1$ , OR  $x = 2$  IF TERMS CANCEL

At  $x = 1$  every power is 1, so the check is a sum of the coefficients — the quickest arithmetic there is. If it happens to give 0 on both sides by accident, try  $x = 2$  as well.

02

Worked examples

EXAMPLE 1 Compute  $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$ .

1

A plus: signs unchanged.

2

$3x^2 - 2x + 5 + x^2 + 4x - 1$

3

Collect by degree.

4

$= 4x^2 + 2x + 4$   
Check at  $x = 1$ :  $10 = 10 \checkmark$

ANSWER

$4x^2 + 2x + 4$

**EXAMPLE 2** Compute  $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$ .

1 A minus: every sign in the second bracket changes.

2  $3x^2 - 2x + 5 - x^2 - 4x + 1$

3 Collect by degree.

4  $= 2x^2 - 6x + 6$

Check at  $x = 1$ :  $2 = 2$  ✓

**ANSWER**

$$2x^2 - 6x + 6$$

**EXAMPLE 3** Simplify  $2(x^2 - x) - 3(x^2 + x)$ .

1 Expand:  $2x^2 - 2x$  and  $-3x^2 - 3x$ .

2  $2x^2 - 2x - 3x^2 - 3x$

3  $x^2$ :  $-x^2$ ;  $x$ :  $-5x$ .

4  $= -x^2 - 5x$

**ANSWER**

$$-x^2 - 5x$$

### 03 Interactive model

Do one subtraction wrongly on purpose — changing only the first sign — and check it at  $x = 1$ ; the check fails, and the class sees why the habit matters.

Checking a simplification

$$(3x^2 - 2x + 1) - (x^2 - 5x + 4) = 2x^2 + 3x - 3$$

x 2

value 11

Both forms give the same value at every  $x$ . That is what makes the simplification correct — and it is how to check one.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH               | O‘ZBEKCHA                 | РУССКИЙ                   |
|-----------------------|---------------------------|---------------------------|
| Sum of polynomials    | Ko‘phadlar yig‘indisi     | Сумма многочленов         |
| Difference            | Ayirma                    | Разность                  |
| To remove brackets    | Qavslarni ochish          | Раскрыть скобки           |
| Like terms            | O‘xshash hadlar           | Подобные члены            |
| Opposite polynomial   | Qarama-qarshi ko‘phad     | Противоположный многочлен |
| Check by substitution | O‘rniga qo‘yib tekshirish | Проверка подстановкой     |
| Standard form         | Standart ko‘rinish        | Стандартный вид           |
| Degree                | Daraja                    | Степень                   |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

A plus before a bracket:

changes all signs

changes none

changes the first

removes the terms

A minus before a bracket:

changes all signs

changes none

changes the first

doubles the terms

$(a + b) - (a - b)$  equals:

$2a$

$2b$

$0$

$2a + 2b$

$(3x^2 - 2x + 5) - (x^2 + 4x - 1)$  equals:

The best value for a quick check:

Adding two polynomials of degree 2 gives degree:

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $(a + b) + (a - b)$

ANSWER  $2a$

2.  $(a + b) - (a - b)$

ANSWER  $2b$

3.  $(x + 2) + (x + 3)$

ANSWER  $2x + 5$

4.  $(x + 2) - (x + 3)$

ANSWER  $-1$

5.  $(2x - 1) + (3x + 4)$

ANSWER  $5x + 3$

6.  $(2x - 1) - (3x + 4)$

ANSWER  $-x - 5$

7. A minus before a bracket

ANSWER Changes every sign

Medium · the standard question

7 PROBLEMS

1.  $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$

ANSWER  $4x^2 + 2x + 4$

2.  $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$

ANSWER  $2x^2 - 6x + 6$

3.  $(2x - 3) + (x + 1) - (x - 4)$

ANSWER  $2x + 2$

4.  $2(x^2 - x) - 3(x^2 + x)$

ANSWER  $-x^2 - 5x$

5.  $(a^2 + ab) - (ab - b^2)$

ANSWER  $a^2 + b^2$

6.  $(5m - 2n) - (3m - 7n)$

ANSWER  $2m + 5n$

7. Check  $4x^2 + 2x + 4$  at  $x = 1$

ANSWER  $10$

### Hard · stretch and reason

7 PROBLEMS

1.  $(x^3 - 2x^2 + x) - (x^3 + x^2 - 3x)$

ANSWER  $-3x^2 + 4x$

2.  $3(2a - b) - 2(a - 3b) + (a - b)$

ANSWER  $5a + 2b$

3. Subtract  $x^2 - 3x + 2$  from  $2x^2 + x - 5$

ANSWER  $x^2 + 4x - 7$

4. What must be added to  $3x - 4$  to give  $5x + 1$ ?

ANSWER  $2x + 5$

5.  $(a + b + c) - (a - b + c)$

ANSWER  $2b$

6. The degree of  $(x^2 + x) - (x^2 - x)$

ANSWER  $1$

7. Simplify  $(2x^2 - 3) + (3 - 2x^2)$

ANSWER  $0$

## 07 Homework

### Homework — 5 tasks

Write the bracket-free line in full, then collect, then check at  $x = 1$ .

1. Compute  $(4x^2 + x - 3) + (2x^2 - 5x + 7)$ .
2. Compute  $(4x^2 + x - 3) - (2x^2 - 5x + 7)$ .
3. Simplify  $3(a - 2b) - 2(2a - b)$ .
4. Subtract  $2y^2 - y + 4$  from  $5y^2 + 3y - 1$ .
5. What must be added to  $7m - 2$  to give  $4m + 5$ ?



# Multiplication of polynomials

Every term of the first times every term of the second — and a rectangle explains why.

LESSON 73–74

2 × 40 MIN

MATEMATIKA 7, §34

S8 2.3

LESSON CLOCK

13'

23'

22'

18'

4'

|                  |        |
|------------------|--------|
| By a monomial    | 13 min |
| Two binomials    | 23 min |
| The area picture | 22 min |
| Longer products  | 18 min |
| Homework         | 4 min  |

LEARNING OBJECTIVES

- Multiply a polynomial by a monomial.
- Multiply two binomials and collect the result.
- Multiply a binomial by a trinomial.
- Use an area diagram to justify the expansion.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 205–211        |
| Cambridge | Stage 8, pp. 18–24 |
| Workbook  | Exercise 2.3       |

01

## Explanation

### By a monomial

$$k(a + b + c) = ka + kb + kc$$

Multiply every term — including the last, and including the signs.

| PRODUCT             | EXPANDED            |
|---------------------|---------------------|
| $3x(x - 4)$         | $3x^2 - 12x$        |
| $-2a(a^2 - 3a + 1)$ | $-2a^3 + 6a^2 - 2a$ |
| $xy(x + y)$         | $x^2y + xy^2$       |

THE LAST TERM IS THE ONE MOST OFTEN FORGOTTEN

In  $-2a(a^2 - 3a + 1)$  the third term gives  $-2a$ . Counting the terms before and after expanding catches the omission.

### Two binomials

$$(a + b)(c + d) = ac + ad + bc + bd$$

Four products, one for each pair. Then collect any like terms.

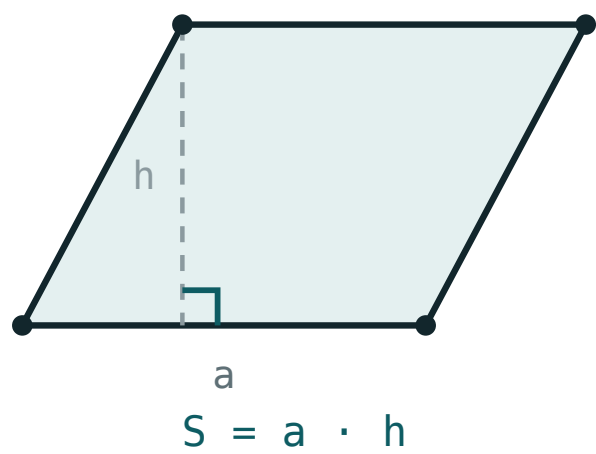
| PRODUCT           | FOUR TERMS           | COLLECTED       |
|-------------------|----------------------|-----------------|
| $(x + 2)(x + 3)$  | $x^2 + 3x + 2x + 6$  | $x^2 + 5x + 6$  |
| $(x - 2)(x + 5)$  | $x^2 + 5x - 2x - 10$ | $x^2 + 3x - 10$ |
| $(2a - 1)(a + 4)$ | $2a^2 + 8a - a - 4$  | $2a^2 + 7a - 4$ |
| $(x - 3)(x + 3)$  | $x^2 + 3x - 3x - 9$  | $x^2 - 9$       |

THE LAST ROW IS WORTH REMEMBERING

The middle terms cancel and only  $x^2 - 9$  survives. That pattern — the difference of two squares — is the subject of a whole lesson in Quarter III.

The area picture

A rectangle of sides  $a + b$  and  $c + d$  splits into four smaller rectangles.



A rectangle cut into four — the four products of the expansion.

|     | $C$  | $D$  |
|-----|------|------|
| $a$ | $ac$ | $ad$ |
| $b$ | $bc$ | $bd$ |

The whole area is the sum of the four parts — which is exactly the expansion. The grid also makes it impossible to miss a product.

USE THE GRID WHEN THERE ARE MORE THAN FOUR PRODUCTS

For a binomial times a trinomial there are six, and a  $2 \times 3$  grid keeps them all in view.

Longer products

**Example.**  $(x + 2)(x^2 - 3x + 1)$ . Six products:

|      | $X^2$  | $-3X$   | $+1$ |
|------|--------|---------|------|
| $x$  | $x^3$  | $-3x^2$ | $x$  |
| $+2$ | $2x^2$ | $-6x$   | $2$  |

$$x^3 - 3x^2 + 2x^2 + x - 6x + 2 = x^3 - x^2 - 5x + 2$$

THE DEGREE OF A PRODUCT IS THE SUM OF THE DEGREES

Here  $1 + 2 = 3$ , and the answer is cubic. Checking the degree first tells you what to expect before any arithmetic.

## 02 Worked examples

**EXAMPLE 1** Expand  $-2a(a^2 - 3a + 1)$ .

①  $-2a \times a^2 = -2a^3$

②  $-2a \times (-3a) = 6a^2$   
Two minuses.

③  $-2a \times 1 = -2a$

④  $= -2a^3 + 6a^2 - 2a$   
Three terms in, three out.

ANSWER

$$-2a^3 + 6a^2 - 2a$$

**EXAMPLE 2** Expand  $(x - 2)(x + 5)$ .

①  $x \cdot x = x^2, x \cdot 5 = 5x.$

②  $-2 \cdot x = -2x, -2 \cdot 5 = -10.$

③  $x^2 + 5x - 2x - 10$

④  $= x^2 + 3x - 10$

ANSWER

$$x^2 + 3x - 10$$

**EXAMPLE 3** Expand  $(x + 2)(x^2 - 3x + 1)$ .

①  $x$  row:  $x^3 - 3x^2 + x$ .

②  $+2$  row:  $2x^2 - 6x + 2$ .

③ Add and collect.

④  $= x^3 - x^2 - 5x + 2$   
Degree 3 ✓

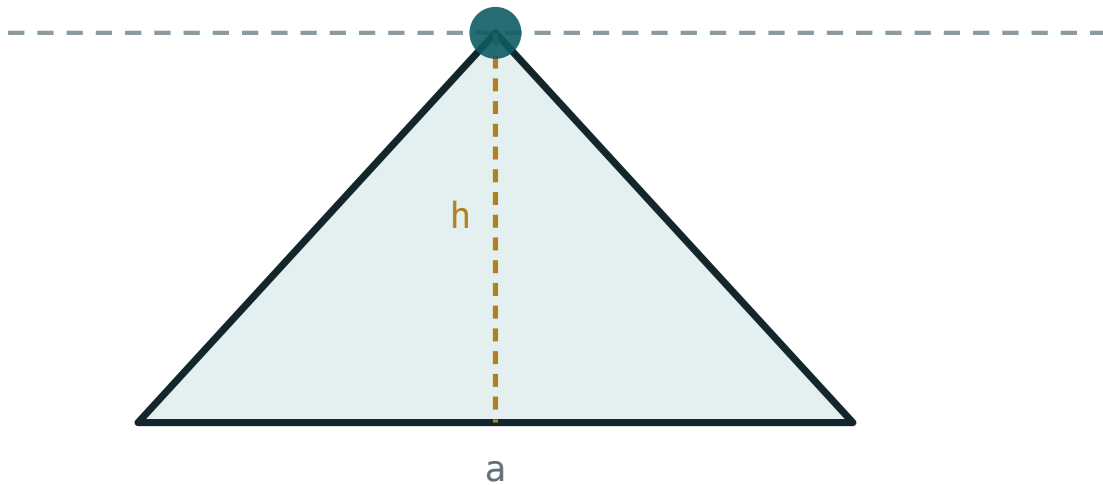
**ANSWER**

$$x^3 - x^2 - 5x + 2$$

**03** **Interactive model**

Draw the rectangle for  $(x + 2)(x + 3)$  with its four parts labelled; the expansion is read off the picture before any rule is stated.

## Expanding as an area

base  $a$  **220**height  $h$  **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**
**04** Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH         | O'ZBEKCHA     | РУССКИЙ             |
|-----------------|---------------|---------------------|
| To multiply out | Ko'paytirish  | Умножение           |
| To expand       | Yoyish        | Раскрыть            |
| Binomial        | Ikki had      | Двучлен             |
| Trinomial       | Uch had       | Трёхчлен            |
| Every term      | Har bir had   | Каждый член         |
| Area diagram    | Yuza chizmasi | Схема площадей      |
| To collect      | Ixchamlash    | Приведение подобных |
| Product         | Ko'paytma     | Произведение        |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$3x(x - 4)$  equals:

$(a + b)(c + d)$  gives how many products?

$(x + 2)(x + 3)$  equals:



$(x - 3)(x + 3)$  equals:

A binomial times a trinomial gives:

The degree of a product is:

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

**Easy · warm the idea up**

7 PROBLEMS

1.  $3x(x - 4)$

ANSWER  $3x^2 - 12x$

2.  $2a(a + 5)$

ANSWER  $2a^2 + 10a$

3.  $xy(x + y)$

ANSWER  $x^2y + xy^2$

4.  $(x + 2)(x + 3)$

ANSWER  $x^2 + 5x + 6$

5.  $(x + 1)(x + 4)$

ANSWER  $x^2 + 5x + 4$

6.  $(x - 3)(x + 3)$

ANSWER  $x^2 - 9$

7.  $(x + 5)(x - 5)$

ANSWER  $x^2 - 25$

**Medium · the standard question**

7 PROBLEMS

1.  $-2a(a^2 - 3a + 1)$

ANSWER  $-2a^3 + 6a^2 - 2a$

2.  $(x - 2)(x + 5)$

ANSWER  $x^2 + 3x - 10$

3.  $(2a - 1)(a + 4)$

ANSWER  $2a^2 + 7a - 4$

4.  $(3x + 2)(x - 1)$

ANSWER  $3x^2 - x - 2$

5.  $(x + 2)(x^2 - 3x + 1)$

ANSWER  $x^3 - x^2 - 5x + 2$

6.  $(a - b)(a + b)$

ANSWER  $a^2 - b^2$

7.  $(2x - 3)(2x + 3)$

ANSWER  $4x^2 - 9$

**Hard · stretch and reason****7 PROBLEMS**

1.  $(x + 1)(x + 2)(x + 3)$

ANSWER  $x^3 + 6x^2 + 11x + 6$

2.  $(a + b)^2$

ANSWER  $a^2 + 2ab + b^2$

3.  $(a - b)^2$

ANSWER  $a^2 - 2ab + b^2$

4.  $(2x - 1)(x^2 + x - 3)$

ANSWER  $2x^3 + x^2 - 7x + 3$

5. A rectangle  $(x + 3)$  by  $(x + 5)$ : its area

ANSWER  $x^2 + 8x + 15$

6. Its perimeter

ANSWER  $4x + 16$

7.  $(x + a)(x + b)$  in general

ANSWER  $x^2 + (a + b)x + ab$

**07 Homework****Homework — 5 tasks**

Count the products you expect before expanding, and again afterwards.

1. Expand  $4x(2x - 3)$  and  $-3a(a^2 + 2a - 5)$ .
2. Expand  $(x + 4)(x + 6)$  and  $(x - 4)(x + 6)$ .
3. Expand  $(3a + 2)(2a - 5)$ .
4. Expand  $(x - 1)(x^2 + 2x + 3)$ .
5. A rectangle has sides  $x + 2$  and  $x + 7$ . Find its area and perimeter.

# Division of polynomials

Dividing by a monomial — term by term, and only when it goes exactly.

LESSON 75–76

2 × 40 MIN

MATEMATIKA 7, §35

S8 2.3

LESSON CLOCK

13'

23'

22'

18'

4'

|                            |        |
|----------------------------|--------|
| Term by term               | 13 min |
| When it works              | 23 min |
| Reversing a multiplication | 22 min |
| Simplifying fractions      | 18 min |
| Homework                   | 4 min  |

LEARNING OBJECTIVES

- Divide a polynomial by a monomial, term by term.
- Decide when the quotient is again a polynomial.
- Use division to reverse a multiplication.
- Simplify an algebraic fraction whose denominator is a monomial.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 212–217        |
| Cambridge | Stage 8, pp. 18–24 |
| Workbook  | Exercise 2.3       |

01

## Explanation

### Term by term

$$\frac{a + b + c}{k} = \frac{a}{k} + \frac{b}{k} + \frac{c}{k}$$

Divide each term of the polynomial by the monomial.

| QUOTIENT                         | TERM BY TERM                      | ANSWER    |
|----------------------------------|-----------------------------------|-----------|
| $(6x^2 + 9x) \div 3x$            | $\frac{6x^2}{3x} + \frac{9x}{3x}$ | $2x + 3$  |
| $(8a^3 - 4a^2) \div 4a^2$        | $2a - 1$                          | $2a - 1$  |
| $(10x^3y - 15x^2y^2) \div 5x^2y$ | $2x - 3y$                         | $2x - 3y$ |

EVERY TERM IS DIVIDED, INCLUDING THE LAST

$(8a^3 - 4a^2) \div 4a^2$  is  $2a - 1$ , not  $2a$ . The second term gives  $-1$ , and dropping it is the standard slip.

### When it works

The quotient is a polynomial only if the monomial divides **every** term exactly — the coefficients and each letter.

| DIVISION                | POLYNOMIAL? | WHY                         |
|-------------------------|-------------|-----------------------------|
| $(6x^2 + 9x) \div 3x$   | yes         | $3x$ divides both terms     |
| $(6x^2 + 9x) \div 3x^2$ | no          | $3x^2$ does not divide $9x$ |
| $(4a + 6b) \div 2$      | yes         | 2 divides both terms        |
| $(4a + 5b) \div 2$      | no          | 2 does not divide $5b$      |

#### THE DIVISOR MUST BE A COMMON FACTOR

Finding the highest common factor of all the terms tells you at once what the polynomial can be divided by — and that is exactly what factorising will use two lessons from now.

## Reversing a multiplication

Division undoes multiplication, so every division can be checked by multiplying back.

| DIVISION               | ANSWER   | CHECK                       |
|------------------------|----------|-----------------------------|
| $(6x^2 + 9x) \div 3x$  | $2x + 3$ | $3x(2x + 3) = 6x^2 + 9x$ ✓  |
| $(10a^2 - 5a) \div 5a$ | $2a - 1$ | $5a(2a - 1) = 10a^2 - 5a$ ✓ |

#### ONE MULTIPLICATION CHECKS THE WHOLE DIVISION

It takes ten seconds and catches every slip, including the dropped last term. There is no faster check in algebra.

## Simplifying fractions

An algebraic fraction is simplified by cancelling the common factors of numerator and denominator.

| FRACTION               | SIMPLIFIED     |
|------------------------|----------------|
| $\frac{12a^3}{4a}$     | $3a^2$         |
| $\frac{6x^2 + 9x}{3x}$ | $2x + 3$       |
| $\frac{5ab}{10a^2}$    | $\frac{b}{2a}$ |

CANCEL FACTORS, NEVER TERMS

In  $\frac{6x^2 + 9x}{3x}$  the whole numerator is divided. Cancelling the  $x$  of  $9x$  against the denominator alone, and leaving the  $6x^2$  untouched, is wrong — the division applies to every term or to none.

02

## Worked examples

---

**EXAMPLE 1** Compute  $(6x^2 + 9x) \div 3x$ .

1

$$\frac{6x^2}{3x} = 2x$$

---

2

$$\frac{9x}{3x} = 3$$

---

3

$$= 2x + 3$$

---

4

Check:  $3x(2x + 3) = 6x^2 + 9x$  ✓

ANSWER

$$2x + 3$$

**EXAMPLE 2** Compute  $(10x^3y - 15x^2y^2) \div 5x^2y$ .

① First term:  $\frac{10x^3y}{5x^2y} = 2x$ .

② Second:  $\frac{-15x^2y^2}{5x^2y} = -3y$ .

③  $= 2x - 3y$

④ Check:  $5x^2y(2x - 3y) = 10x^3y - 15x^2y^2 \checkmark$

**ANSWER**

$$2x - 3y$$

**EXAMPLE 3** Is  $(6x^2 + 9x) \div 3x^2$  a polynomial?

① First term:  $\frac{6x^2}{3x^2} = 2 \checkmark$

② Second:  $\frac{9x}{3x^2} = \frac{3}{x}$ .

③ That is not a monomial.

④ No — the quotient is not a polynomial.

**ANSWER**

No

### 03 Interactive model

Ask the class to check every division by multiplying back, and mark only the checks for one lesson; the accuracy of the divisions rises immediately.



**Cancelling common factors**

$$\frac{6x^2 + 9x}{3x}$$

①  $\frac{6x^2}{3x} = 2x$

The  
first  
term.

②  $\frac{9x}{3x} = 3$

The  
second  
— it  
does  
not  
vanish.

③  $2x + 3$

Check:

$$\begin{aligned} &3x(2x \\ &+ 3) \\ &= \\ &6x^2 \\ &+ 9x \\ &\checkmark \end{aligned}$$

**04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA           | РУССКИЙ              |
|--------------------|---------------------|----------------------|
| Division           | Bo'lish             | Деление              |
| Quotient           | Bo'linma            | Частное              |
| Term by term       | Hadma-had           | Почленно             |
| Exactly divisible  | Butun bo'linadi     | Делится нацело       |
| Common factor      | Umumiy ko'paytuvchi | Общий множитель      |
| To reverse         | Teskari amal        | Обратное действие    |
| Algebraic fraction | Algebraik kasr      | Алгебраическая дробь |
| Check              | Tekshirish          | Проверка             |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

To divide a polynomial by a monomial:

divide the first term

divide every term

multiply

cancel one letter

$(8a^3 - 4a^2) \div 4a^2$  equals:

$2a$

$2a - 1$

$2a^3$

$2a - 4$

$(6x^2 + 9x) \div 3x$  equals:

$2x$

$2x + 3$

$2x + 9$

$2x^2 + 3$

$(6x^2 + 9x) \div 3x^2$  is:

a polynomial

not a polynomial

a monomial

zero

A division is checked by:

dividing again

multiplying back

estimating

nothing

In an algebraic fraction you may cancel:

terms

factors

anything

signs

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $(6x^2 + 9x) \div 3x$

ANSWER  $2x + 3$ 

2.  $(8a^3 - 4a^2) \div 4a^2$

ANSWER  $2a - 1$ 

3.  $(4a + 6b) \div 2$

ANSWER  $2a + 3b$ 

4. 
$$\frac{12a^3}{4a}$$

ANSWER  $3a^2$ 

5.  $(10a^2 - 5a) \div 5a$

ANSWER  $2a - 1$ 

6.  $(x^2 + x) \div x$

ANSWER  $x + 1$ 

7.  $(9y^3 + 3y) \div 3y$

ANSWER  $3y^2 + 1$ **Medium · the standard question**

7 PROBLEMS

1.  $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER  $2x - 3y$

2.  $(12a^2b + 8ab^2) \div 4ab$

ANSWER  $3a + 2b$

3.  $\frac{5ab}{10a^2}$

ANSWER  $\frac{b}{2a}$

4. Is  $(6x^2 + 9x) \div 3x^2$  a polynomial?

ANSWER No

5. Is  $(4a + 5b) \div 2$  a polynomial?

ANSWER No

6. Check  $2x + 3$  by multiplying by  $3x$

ANSWER  $6x^2 + 9x$

7.  $(a^3 - a^2) \div a^2$

ANSWER  $a - 1$

**Hard · stretch and reason**

7 PROBLEMS

1.  $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER  $3x^2y - 2xy^2 + 1$

2. 
$$\frac{4a^2b - 6ab^2}{2ab}$$

ANSWER  $2a - 3b$

3. What must  $3x$  be multiplied by to give  $12x^3 - 9x^2$ ?

ANSWER  $4x^2 - 3x$

4. 
$$\frac{x^2 - x}{x}$$

ANSWER  $x - 1$

5. The HCF of  $6x^2$  and  $9x$

ANSWER  $3x$

6. Divide  $8a^3b - 12a^2b^2 + 4ab$  by its HCF

ANSWER  $2a^2 - 3ab + 1$

7. Why must the divisor be a common factor?

ANSWER Otherwise some term gives a fraction

## 07 Homework

### Homework — 5 tasks

Check every division by multiplying the answer by the divisor.

1. Compute  $(12x^2 - 8x) \div 4x$ .

2. Compute  $(9a^3b - 6a^2b^2) \div 3a^2b$ .

3. Simplify  $\frac{20m^3}{5m}$ .

4. Is  $(10x^2 + 4x) \div 4x^2$  a polynomial? Explain.

5. Find the HCF of  $8a^2b$  and  $12ab^2$ , and divide their sum by it.



# Factorising a polynomial

Taking out the common factor, and grouping when there is no single one.

LESSON 77–78

2 × 40 MIN

MATEMATIKA 7, §36

S8 2.4

LESSON CLOCK

9'

23'

22'

22'

4'

|                   |        |
|-------------------|--------|
| The idea          | 9 min  |
| The common factor | 23 min |
| Grouping          | 22 min |
| What it is for    | 22 min |
| Homework          | 4 min  |

LEARNING OBJECTIVES

- Find the highest common factor of the terms of a polynomial.
- Take that factor outside a bracket, and check by expanding.
- Factorise a four-term polynomial by grouping.
- Use a factorised form to evaluate an expression quickly.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 218–224        |
| Cambridge | Stage 8, pp. 25–30 |
| Workbook  | Exercise 2.4       |

01

## Explanation

### The idea

Expanding turns a product into a sum. **Factorising** is the same road travelled backwards: it turns a sum into a product.

$$3x(2x + 3) = 6x^2 + 9x \rightarrow 6x^2 + 9x = 3x(2x + 3)$$

| DIRECTION   | START        | FINISH       |
|-------------|--------------|--------------|
| expanding   | $3x(2x + 3)$ | $6x^2 + 9x$  |
| factorising | $6x^2 + 9x$  | $3x(2x + 3)$ |

THE CHECK IS FREE

Every factorisation can be checked by expanding it again. If the expansion does not return the original polynomial exactly, the factorisation is wrong — there is nothing to argue about.

## The common factor

Take the highest common factor of **all** the terms outside the bracket. For the numbers that is their HCF; for each letter it is the lowest power that appears in every term.

| POLYNOMIAL             | HCF     | FACTORISED         |
|------------------------|---------|--------------------|
| $6x^2 + 9x$            | $3x$    | $3x(2x + 3)$       |
| $8a^3 - 4a^2$          | $4a^2$  | $4a^2(2a - 1)$     |
| $10x^3y - 15x^2y^2$    | $5x^2y$ | $5x^2y(2x - 3y)$   |
| $12a^2b + 8ab^2 - 4ab$ | $4ab$   | $4ab(3a + 2b - 1)$ |

A TERM THAT IS ENTIRELY THE FACTOR LEAVES A 1

In  $12a^2b + 8ab^2 - 4ab$  the last term is  $-4ab$ , which is the factor itself; inside the bracket it becomes  $-1$ , not nothing. Losing that  $1$  is the commonest error of the whole topic, and expanding back catches it at once.

## Grouping

When the four terms have no common factor between all of them, split them into two pairs that do, and a common bracket appears.

$$ax + ay + bx + by = a(x + y) + b(x + y) = (x + y)(a + b)$$

| STEP                  | WORKING                   |
|-----------------------|---------------------------|
| the polynomial        | $2x^2 + 6x + ax + 3a$     |
| pair them             | $(2x^2 + 6x) + (ax + 3a)$ |
| factorise each pair   | $2x(x + 3) + a(x + 3)$    |
| the bracket is common | $(x + 3)(2x + a)$         |

A MINUS IN FRONT OF THE SECOND PAIR CHANGES BOTH SIGNS

$x^2 - 3x - 2x + 6$  pairs as  $x(x - 3) - 2(x - 3)$ , giving  $(x - 3)(x - 2)$ . Writing  $-2(x + 3)$  there is a sign slip, and the two brackets then fail to match.

## What it is for

A product is easier to evaluate, easier to cancel and — next year — the only way to solve an equation.

| USE          | EXAMPLE                     | WHY THE PRODUCT HELPS            |
|--------------|-----------------------------|----------------------------------|
| evaluating   | $37 \cdot 24 + 37 \cdot 76$ | $37(24 + 76) = 3700$             |
| cancelling   | $\frac{6x^2 + 9x}{3x}$      | $\frac{3x(2x + 3)}{3x} = 2x + 3$ |
| divisibility | $n^2 + n$                   | $n(n + 1)$                       |

### WHY $n^2 + n$ IS ALWAYS EVEN

Factorised it is  $n(n + 1)$  — the product of two consecutive whole numbers, one of which must be even. In the unfactorised form the fact is invisible; that is the whole value of a product.

## 02 Worked examples

### EXAMPLE 1 Factorise $12a^2b + 8ab^2 - 4ab$ .

- ① HCF of 12, 8, 4 is 4.
- ② Every term has  $a$  and  $b$ , so the HCF is  $4ab$ .
- ③  $= 4ab(3a + 2b - 1)$   
The last term leaves  $-1$ .
- ④ Check:  $4ab(3a + 2b - 1) = 12a^2b + 8ab^2 - 4ab$  ✓

#### ANSWER

$$4ab(3a + 2b - 1)$$

**EXAMPLE 2** Factorise  $2x^2 + 6x + ax + 3a$  by grouping.

$$① \quad (2x^2 + 6x) + (ax + 3a)$$

$$② \quad 2x(x + 3) + a(x + 3)$$

The same bracket in both.

$$③ \quad = (x + 3)(2x + a)$$

$$④ \quad \text{Check: } (x + 3)(2x + a) = 2x^2 + ax + 6x + 3a \quad \checkmark$$

**ANSWER**

$$(x + 3)(2x + a)$$

**EXAMPLE 3** Evaluate  $37 \cdot 24 + 37 \cdot 76$  without a calculator.

$$① \quad \text{Both terms contain 37.}$$

$$② \quad = 37(24 + 76)$$

$$③ \quad = 37 \cdot 100$$

$$④ \quad = 3700$$

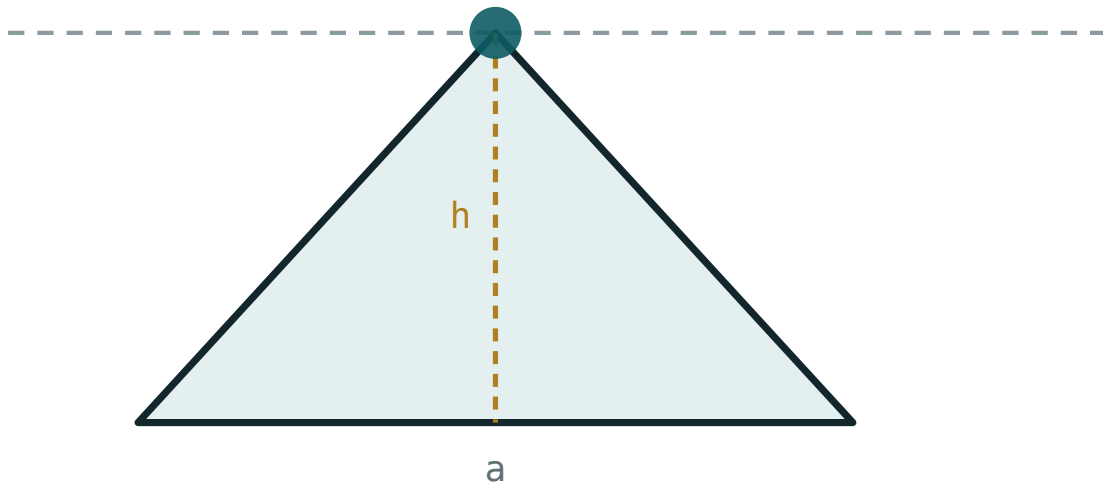
**ANSWER**

$$3700$$

**03 Interactive model**

Give the class the factorised answers first and ask them to expand; when they then meet the sums, the factor is already familiar and the missing 1 stops appearing.

### A common factor as one side of a rectangle

base  $a$  **220**height  $h$  **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH                     | O'ZBEKCHA                    | РУССКИЙ                   |
|-----------------------------|------------------------------|---------------------------|
| To factorise                | Ko'paytuvchilarga ajratish   | Разложить на множители    |
| Factor                      | Ko'paytuvchi                 | Множитель                 |
| Highest common factor       | Eng katta umumiy bo'luvchi   | Наибольший общий делитель |
| To take outside the bracket | Qavsdan tashqariga chiqarish | Вынести за скобку         |
| Grouping                    | Guruhlash                    | Группировка               |
| Product                     | Ko'paytma                    | Произведение              |
| Sum                         | Yig'indi                     | Сумма                     |
| Check by expanding          | Qavsni ochib tekshirish      | Проверка раскрытием       |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Factorising turns a sum into:

a difference

a product

a quotient

a power

$6x^2 + 9x$  factorises to:

$3x(2x + 3)$

$3x(2x + 9)$

$x(6x + 9)$

$3(2x^2 + 3x)$

In  $12a^2b + 8ab^2 - 4ab$  the last term leaves:

0

-1

$-4ab$

1

Grouping is used when:

there are two terms

no factor is common to all

the terms are like

there is a power

$ax + ay + bx + by$  equals:

$(x + y)(a + b)$

$(a + x)(b + y)$

$ab(x + y)$

$(a + b)^2$

A factorisation is checked by:

dividing

expanding

substituting 0

nothing

## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1.  $4x + 8$

ANSWER  $4(x + 2)$

2.  $6x^2 + 9x$

ANSWER  $3x(2x + 3)$

3.  $8a^3 - 4a^2$

ANSWER  $4a^2(2a - 1)$

4.  $5x - 15$

ANSWER  $5(x - 3)$

5.  $x^2 + x$

ANSWER  $x(x + 1)$

6.  $9y^3 + 3y$

ANSWER  $3y(3y^2 + 1)$

7.  $10ab - 5a$

ANSWER  $5a(2b - 1)$

**Medium · the standard question**

7 PROBLEMS

1.  $10x^3y - 15x^2y^2$

ANSWER  $5x^2y(2x - 3y)$

2.  $12a^2b + 8ab^2 - 4ab$

ANSWER  $4ab(3a + 2b - 1)$

3.  $ax + ay + bx + by$

ANSWER  $(x + y)(a + b)$

4.  $2x^2 + 6x + ax + 3a$

ANSWER  $(x + 3)(2x + a)$

5.  $x^2 - 3x - 2x + 6$

ANSWER  $(x - 3)(x - 2)$

6.  $37 \cdot 24 + 37 \cdot 76$

ANSWER  $3700$

7.  $14m^2n - 21mn^2$

ANSWER  $7mn(2m - 3n)$

**Hard · stretch and reason**

7 PROBLEMS

1.  $15x^4y^2 - 10x^3y^3 + 5x^2y$

ANSWER  $5x^2y(3x^2y - 2xy^2 + 1)$

2.  $3a(x - 2) + 5(x - 2)$

ANSWER  $(x - 2)(3a + 5)$

3.  $x(a - b) - y(b - a)$

ANSWER  $(a - b)(x + y)$

4. Show that  $n^2 + n$  is always even

ANSWER  $n(n + 1)$  — one of them is even

5.  $56 \cdot 19 + 56 \cdot 81$

ANSWER 5600

6. Simplify  $\frac{4a^2b - 6ab^2}{2ab}$  by factorising first

ANSWER  $2a - 3b$

7. Factorise  $x^3 + x^2 + x + 1$

ANSWER  $(x + 1)(x^2 + 1)$

**07 Homework****Homework — 5 tasks**

Expand every answer back; a factorisation that does not return the original is not an answer.

- Factorise  $12x^2 - 8x$  and  $9a^3b - 6a^2b^2$ .
- Factorise  $6m^2n + 9mn^2 - 3mn$ , and say what the last term leaves.
- Factorise  $3x^2 + 12x + ax + 4a$  by grouping.
- Evaluate  $48 \cdot 37 + 48 \cdot 63$  by factorising.

5. Explain, using a factorisation, why  $n^2 - n$  is always even.

# Control work 5, and work on the mistakes

Expressions, powers, monomials and polynomials in one paper — Quarter II closed.

LESSON 79–80

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 5

S8 2 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

|                       |        |
|-----------------------|--------|
| Instructions          | 3 min  |
| The paper             | 40 min |
| Answers               | 12 min |
| Diagnosis and rewrite | 20 min |
| The map               | 5 min  |

LEARNING OBJECTIVES

- Apply the laws of indices without confusing them with the laws of arithmetic.
- Add, subtract, multiply and divide polynomials accurately under time.
- Factorise by the common factor and by grouping.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

|           |                    |
|-----------|--------------------|
| National  | pp. 152–224        |
| Cambridge | Stage 8, pp. 12–30 |
| Workbook  | Control paper M5   |

01

## Explanation

### The paper — 25 marks, 40 minutes

| Q | TASK                                                            | MARKS | FROM   |
|---|-----------------------------------------------------------------|-------|--------|
| 1 | Simplify $a^5 \cdot a^3$ , $a^8 \div a^3$ , $(a^2)^4$ and $a^0$ | 4     | L63–64 |
| 2 | Multiply the monomials $3x^2y$ and $-4xy^3$                     | 3     | L66–67 |
| 3 | Collect $5a^2 - 3a + 2 - 2a^2 + 7a - 5$                         | 4     | L69–70 |
| 4 | Expand $(x + 4)(x - 3)$ and $-3a(a^2 + 2a - 5)$                 | 5     | L73–74 |
| 5 | Divide $(10x^3y - 15x^2y^2) \div 5x^2y$                         | 4     | L75–76 |
| 6 | Factorise $12a^2b + 8ab^2 - 4ab$ and $2x^2 + 6x + ax + 3a$      | 5     | L77–78 |

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for  $a^0 = 1$  alone; Q2 one for the sign; Q4 two for the four products of the binomial pair and one for the sign of the last term; Q6 one for the  $-1$  left by the third term and two for the common bracket in the grouping.

## Naming the slip

| SLIP                            | WHAT IT LOOKS LIKE                | THE FIX                |
|---------------------------------|-----------------------------------|------------------------|
| indices multiplied              | $a^5 \cdot a^3 = a^{15}$          | $a^8$ — add them       |
| zero index misread              | $a^0 = 0$                         | $a^0 = 1$              |
| sign of a product               | $3x^2y \cdot (-4xy^3) = 12x^3y^4$ | $-12x^3y^4$            |
| unlike terms collected          | $5a^2 - 3a = 2a$                  | they are not like      |
| only two products               | $(x + 4)(x - 3) = x^2 - 12$       | all four, then collect |
| last term of a division dropped | $2x$                              | $2x - 3y$              |
| the 1 lost when factorising     | $4ab(3a + 2b)$                    | $4ab(3a + 2b - 1)$     |

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

## Quarter II as one map

| BLOCK          | THE SENTENCE                                                       |
|----------------|--------------------------------------------------------------------|
| expressions    | a letter stands for a number, so the laws of arithmetic still hold |
| powers         | an index counts the factors, so multiplying adds the indices       |
| monomials      | one product of numbers and letters, with a degree                  |
| polynomials    | a sum of monomials; only like terms may be collected               |
| multiplication | every term of the first by every term of the second                |
| division       | term by term, and only when the divisor is a common factor         |
| factorising    | the common factor outside, or grouping in pairs                    |

LOOKING FORWARD

Quarter III opens with geometry again — parallel lines, the angle sum of a triangle and right-angled triangles — and returns to algebra with the abridged multiplication formulae, which are the products of Q4 learned by heart.

## 02 Worked examples

**EXAMPLE 1** Model answer, Q1: the four index tasks.

①  $a^5 \cdot a^3 = a^8$   
Add the indices.

②  $a^8 \div a^3 = a^5$   
Subtract them.

③  $(a^2)^4 = a^8$   
Multiply them.

④  $a^0 = 1$   
For every  $a \neq 0$ .

**ANSWER**

$a^8, a^5, a^8, 1$

**EXAMPLE 2** Model answer, Q4: expand  $(x + 4)(x - 3)$ .

$$1 \quad x \cdot x = x^2$$

$$2 \quad x \cdot (-3) = -3x \text{ and } 4 \cdot x = 4x.$$

$$3 \quad 4 \cdot (-3) = -12$$

The sign is the mark.

$$4 \quad = x^2 + x - 12$$

ANSWER

$$x^2 + x - 12$$

**EXAMPLE 3** Model answer, Q6: factorise  $2x^2 + 6x + ax + 3a$ .

$$1 \quad (2x^2 + 6x) + (ax + 3a)$$

$$2 \quad 2x(x + 3) + a(x + 3)$$

$$3 \quad = (x + 3)(2x + a)$$

$$4 \quad \text{Check by expanding: } 2x^2 + ax + 6x + 3a \checkmark$$

ANSWER

$$(x + 3)(2x + a)$$

**03 Interactive model**

Hand back Q4 and Q6 together and ask which is which: expanding and factorising are one skill in two directions, and seeing them side by side fixes it.



## Quarter II in twelve questions

 $a^5 \cdot a^3$  equals:

$a^{15}$

$a^8$

$a^2$

$2a^8$

 $a^8 \div a^3$  equals:

$a^5$

$a^{11}$

$a^2$

$a^{24}$

 $(a^2)^4$  equals:

$a^6$

$a^8$

$a^{16}$

$2a^4$

$a^0$  equals:

0

1

$a$

undefined

The degree of  $3x^2y$  is:

2

3

5

1

$3x^2y \cdot (-4xy^3)$  equals:

$12x^3y^4$

$-12x^3y^4$

$-12x^2y^3$

$-7x^3y^4$

Like terms differ only in:

the letters

the coefficient

the degree

the sign of the letter

$5a^2 - 3a + 2 - 2a^2 + 7a - 5$  equals:

$$3a^2 + 4a - 3$$

$$7a^2 + 4a - 3$$

$$3a^2 - 4a + 3$$

$$3a^2 + 10a - 3$$

$(x + 4)(x - 3)$  equals:

$$x^2 - 12$$

$$x^2 + x - 12$$

$$x^2 + 7x - 12$$

$$x^2 - x - 12$$

$(8a^3 - 4a^2) \div 4a^2$  equals:

$$2a$$

$$2a - 1$$

$$2a - 4$$

$$2a^3$$

$12a^2b + 8ab^2 - 4ab$  factorises to:

$$4ab(3a + 2b)$$

$$4ab(3a + 2b - 1)$$

$$ab(12a + 8b - 4)$$

$$4(3a^2b + 2ab^2 - ab)$$

A factorisation is checked by:

dividing

expanding

substituting

nothing

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH      | O'ZBEKCHA                  | РУССКИЙ                |
|--------------|----------------------------|------------------------|
| Control work | Nazorat ishi               | Контрольная работа     |
| Power        | Daraja                     | Степень                |
| Monomial     | Bir had                    | Одночлен               |
| Polynomial   | Ko'phad                    | Многочлен              |
| Like terms   | O'xshash hadlar            | Подобные члены         |
| To expand    | Qavsni ochish              | Раскрыть скобки        |
| To factorise | Ko'paytuvchilarga ajratish | Разложить на множители |
| Diagnosis    | Tashxis                    | Диагностика            |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Q1 is answered with:

the laws of arithmetic

the laws of indices

a calculator

estimation

The sign in Q2 comes from:

the letters

one negative factor

the degree

the coefficients only

In Q3 the terms  $5a^2$  and  $-3a$  are:

like

not like

equal

opposite

Q4 needs how many products for the binomial pair?





Q5 is wrong if:





The commonest slip in Q6 is:





## 06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1.  $a^5 \cdot a^3$

ANSWER  $a^8$ 

2.  $a^8 \div a^3$

ANSWER  $a^5$ 

3.  $(a^2)^4$

ANSWER  $a^8$ 

4.  $a^0$ , for  $a \neq 0$

ANSWER  $1$ 

5.  $3x^2y \cdot (-4xy^3)$

ANSWER  $-12x^3y^4$ 

6.  $5a^2 - 2a^2$

ANSWER  $3a^2$ 

7.  $(x^2 + x) \div x$

ANSWER  $x + 1$ **Medium · the standard question**

7 PROBLEMS

1.  $5a^2 - 3a + 2 - 2a^2 + 7a - 5$

ANSWER  $3a^2 + 4a - 3$

2.  $(x + 4)(x - 3)$

ANSWER  $x^2 + x - 12$

3.  $-3a(a^2 + 2a - 5)$

ANSWER  $-3a^3 - 6a^2 + 15a$

4.  $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER  $2x - 3y$

5.  $12a^2b + 8ab^2 - 4ab$

ANSWER  $4ab(3a + 2b - 1)$

6.  $2x^2 + 6x + ax + 3a$

ANSWER  $(x + 3)(2x + a)$

7. The degree of  $3x^2y$

ANSWER 3

**Hard · stretch and reason****7 PROBLEMS**



1.  $(2x - 1)(x^2 + x - 3)$

ANSWER  $2x^3 + x^2 - 7x + 3$ 

2.  $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER  $3x^2y - 2xy^2 + 1$ 

3.  $x(a - b) - y(b - a)$

ANSWER  $(a - b)(x + y)$ 

4. Simplify  $\frac{6x^2 + 9x}{3x}$

ANSWER  $2x + 3$ 

5. Expand and collect  $(a + b)^2 - (a - b)^2$

ANSWER  $4ab$ 

6.  $48 \cdot 37 + 48 \cdot 63$

ANSWER  $4800$ 

7. Why is  $n^2 + n$  always even?

ANSWER  $n(n + 1)$  — one factor is even**07 Homework****Homework — 5 tasks**

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write out the four laws of indices with one example each.
3. Expand  $(2x - 5)(x + 3)$  and check by substituting  $x = 1$ .
4. Factorise  $9m^2n + 12mn^2 - 3mn$  and expand your answer back.

5. Write one sentence for each of the seven blocks on the Quarter II map.